

A
SEMINAR REPORT
ON

“CHATBOT”

Submitted
To

UDHNA CITIZEN COMMERCE COLLEGE &
S.P.B. COLLEGE OF BUSINESS ADMINISTRATION &
SMT. DIWALIBEN HARJIBHAI GONDALIA COLLEGE OF BCA AND IT

Affiliated
To

VEER NARMAD SOUTH GUJARAT UNIVERSITY, SURAT

As A Partial Fulfillment For The Degree Of
BACHELOR OF COMPUTER APPLICATION (B.C.A.)
T.Y.B.C.A. (SEM. - 6)
ACADEMIC YEAR: 2022-23

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[SURAT]



**Udhna Citizen Commerce College
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Smt. Diwaliben Harjibhai Gondalia College of BCA & I.T.
(Self Financed College Affiliated To Veer Narmad South Gujarat University, Surat)
(Managed by Udhna Academy Education Trust)

(Bachelor of Computer Application)

CERTIFICATE

This is to Certify that Mr. / Ms. **DUBEY ANIKET DEVENDRA**

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Seminar Work entitled as **Chat bot**

*in accordance with technical and theoretical specification for the academic
year 20.22 to 20.23 towards a Partial fulfillment of BCA Programme.*

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1. Introduction



A chatbot is a computer program designed to simulate human conversation through text or voice interactions. These bots can be integrated into various messaging platforms, such as Facebook Messenger, WhatsApp, or website chat windows. Chatbots can provide a range of services, from customer support and sales to entertainment and personal assistance. They use natural language processing (NLP) and machine learning algorithms to understand user queries and provide relevant responses. With the advancement of AI technology, chatbots have become increasingly sophisticated and are now capable of handling complex conversations and tasks. The popularity of chatbots is growing rapidly, as they offer businesses an efficient and cost-effective way to interact with their customers and improve their overall experience.

1.1 Definition of chatbots

A chatbot is a computer program that uses artificial intelligence (AI) and natural language processing (NLP) to simulate human conversation through text or voice interactions. It is designed to understand and respond to user queries in a conversational manner, typically through messaging platforms or website chat windows. Chatbots can be programmed to perform a wide range of tasks, from answering frequently asked questions and providing customer support to booking appointments and executing transactions. They are becoming increasingly popular due to their ability to improve customer experience and engagement while also reducing operational costs for businesses.

Chatbot

1.2 Importance of chatbots in modern business

WHAT IS A CHATBOT

And Why is it Important for Your Business



Chatbots have become increasingly important in modern business due to their ability to provide efficient and effective customer service while reducing operational costs. Here are some key reasons why chatbots are important in modern business:

24/7 Availability: Chatbots can provide 24/7 support to customers, without the need for human operators. This ensures that customers can receive assistance at any time, increasing customer satisfaction and loyalty.

Cost-effective: Chatbots can handle a large volume of customer inquiries simultaneously, reducing the need for human operators and decreasing operational costs for businesses.

Personalization: Chatbots can use AI and NLP to personalize the customer experience, providing tailored responses based on the customer's history and preferences.

Improved efficiency: Chatbots can quickly and accurately respond to customer queries, reducing the response time and improving the efficiency of customer service.

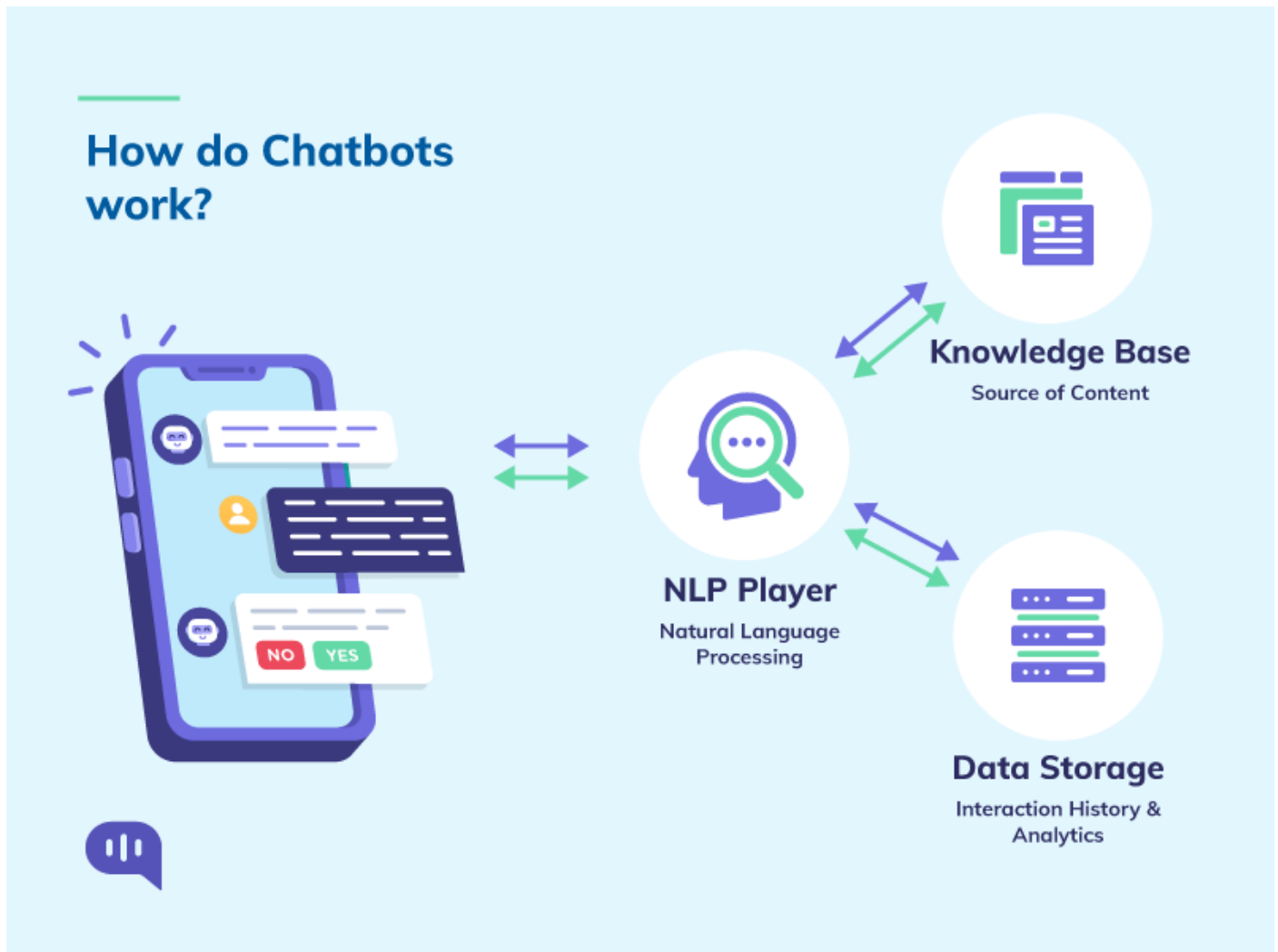
Scalability: Chatbots can handle a large volume of inquiries, making them ideal for businesses that need to scale their customer support operations quickly.

Data collection: Chatbots can collect and analyze data from customer interactions, providing businesses with valuable insights into customer behavior and preferences.

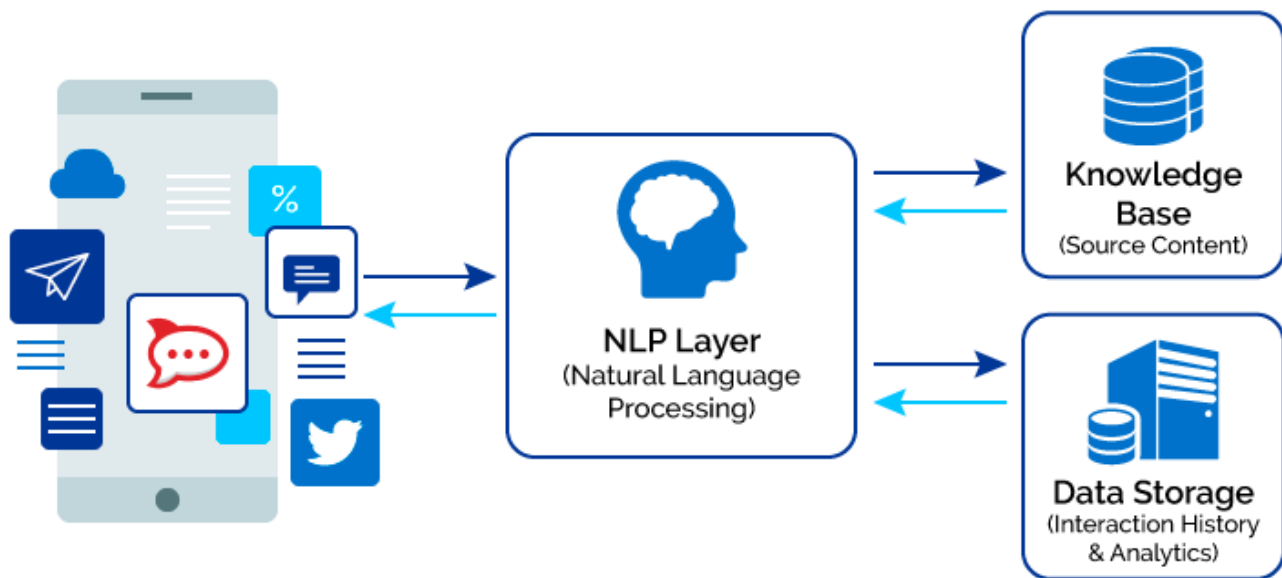
Overall, chatbots have become an essential tool for businesses looking to improve customer experience, increase efficiency, and reduce costs.

Chatbot

2. *How Chatbots Work?*



Chatbot



Chatbots use a combination of natural language processing (NLP) and artificial intelligence (AI) technologies to simulate human conversation and provide responses to user queries. Here's a general overview of how chatbots work:

Understanding the User Query: When a user interacts with a chatbot, the bot analyzes the user's query using NLP technology. NLP helps the chatbot understand the user's intent and extract the relevant information from the user's message.

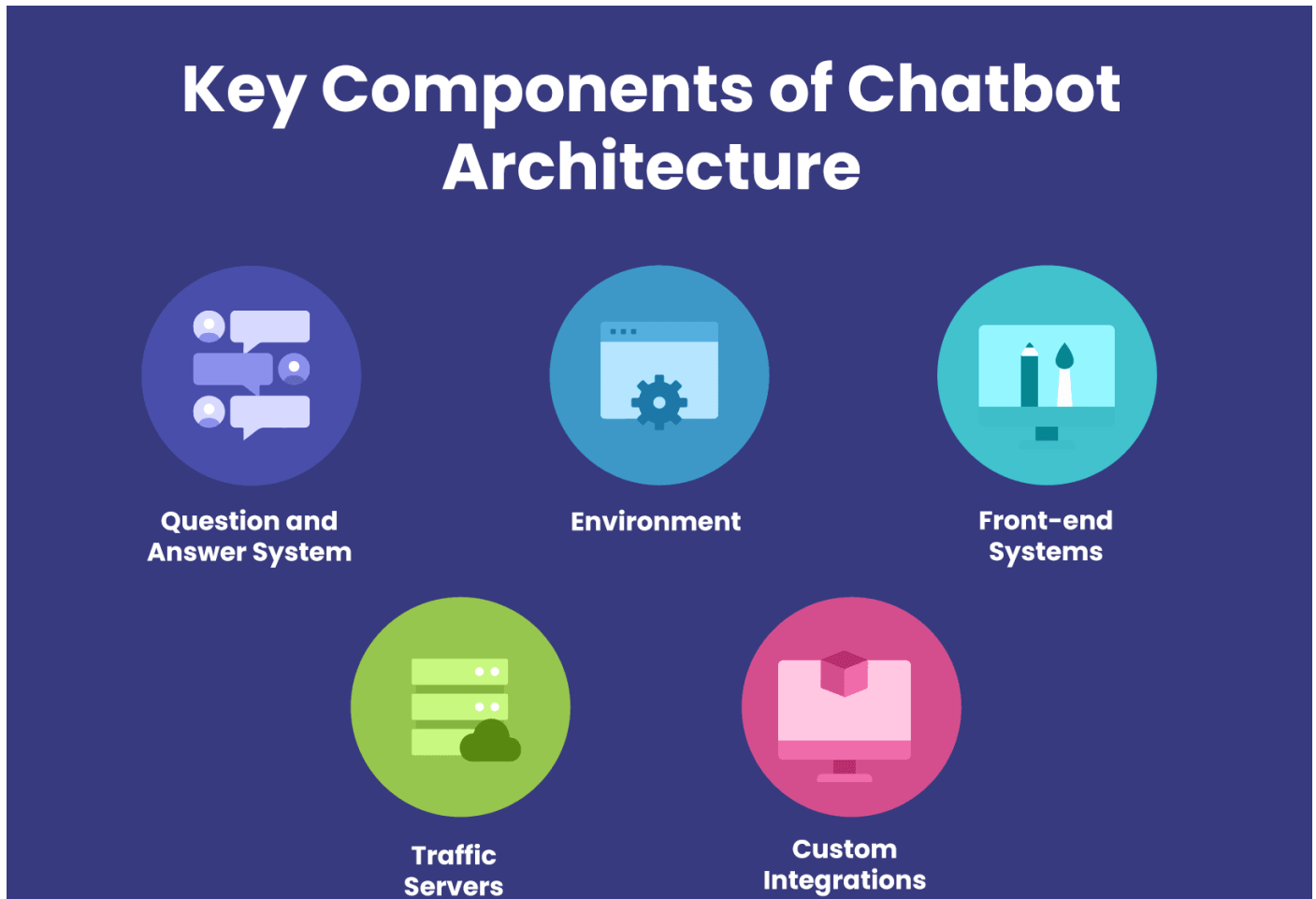
Generating a Response: Once the chatbot has analyzed the user's query, it uses AI algorithms to generate a response. The chatbot may access a database of pre-programmed responses, or it may use machine learning to generate a response on the fly.

Delivering the Response: Once the chatbot has generated a response, it delivers the response to the user through the messaging platform or chat window. The response may be in the form of text, voice, or visual media.

Learning and Improvement: Chatbots use machine learning algorithms to learn from previous interactions and improve their responses over time. This means that as the chatbot interacts with more users, it becomes better at understanding user queries and providing relevant responses.

Overall, chatbots rely on a combination of NLP and AI technologies to simulate human conversation and provide efficient and effective customer service. By continuously learning and improving, chatbots can provide increasingly sophisticated and personalized responses to user queries.

3. Chatbot architecture



The architecture of a chatbot refers to the underlying framework that supports the chatbot's functions and capabilities. There are several components that make up a chatbot's architecture:

User Interface: The user interface is the platform that users interact with to communicate with the chatbot. This could be a messaging platform, such as Facebook Messenger or WhatsApp, or a website chat window.

Natural Language Processing (NLP): The NLP component of a chatbot's architecture is responsible for understanding and interpreting user queries. It analyzes the user's input and extracts the relevant information, such as the user's intent and specific parameters.

Dialog Management: The dialog management component of a chatbot's architecture manages the conversation flow between the chatbot and the user. It determines the appropriate response based on the user's input and previous conversation history.

Business Logic: The business logic component of a chatbot's architecture defines the chatbot's functionality and capabilities. It specifies the tasks that the chatbot can perform and the data sources it can access.

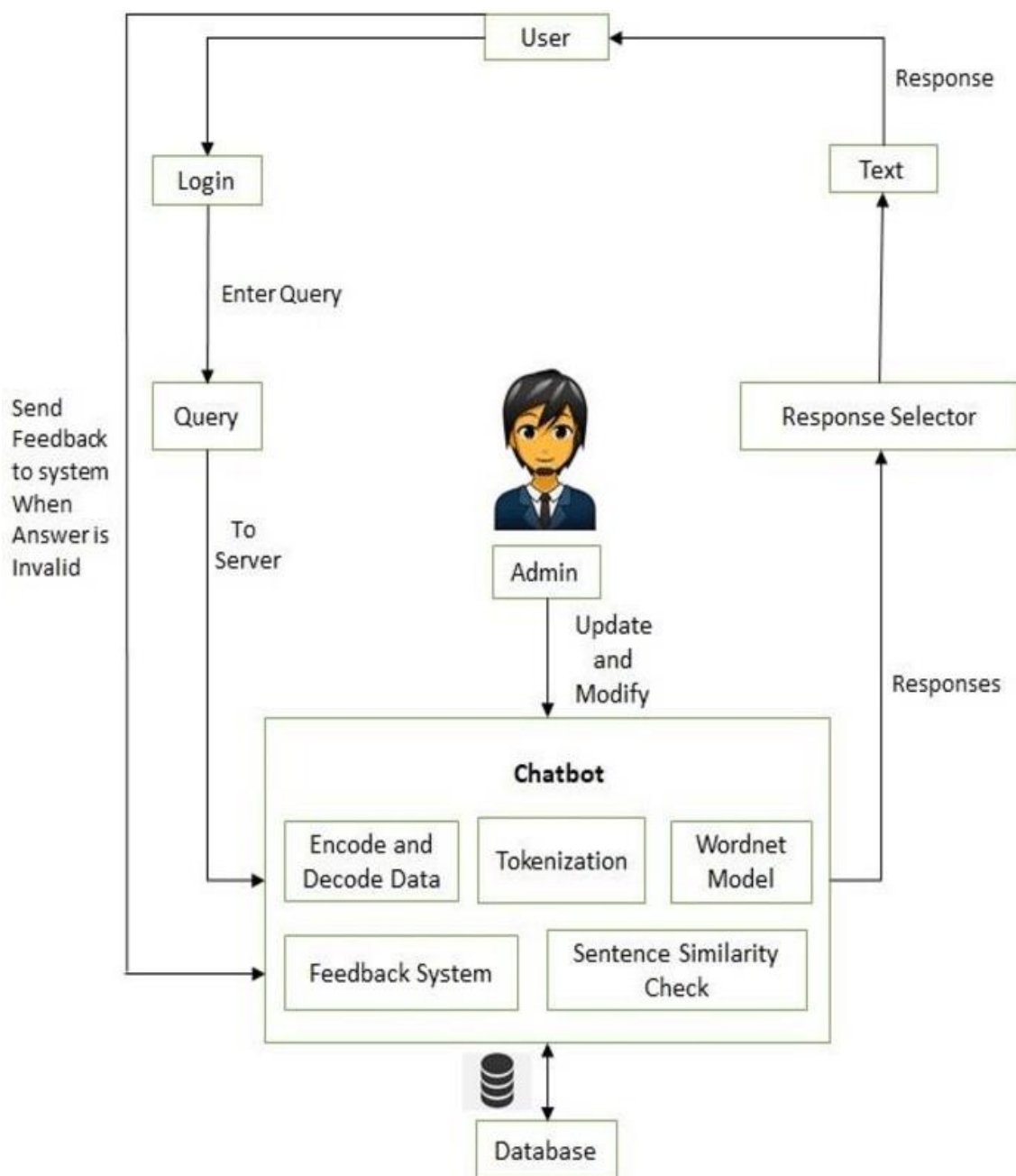


Chatbot

API and Integrations: The API and integrations component of a chatbot's architecture enables the chatbot to interact with external services and data sources, such as payment gateways, customer relationship management (CRM) systems, or databases.

Machine Learning: The machine learning component of a chatbot's architecture enables the chatbot to learn and improve over time. It uses algorithms to analyze user interactions and optimize the chatbot's performance.

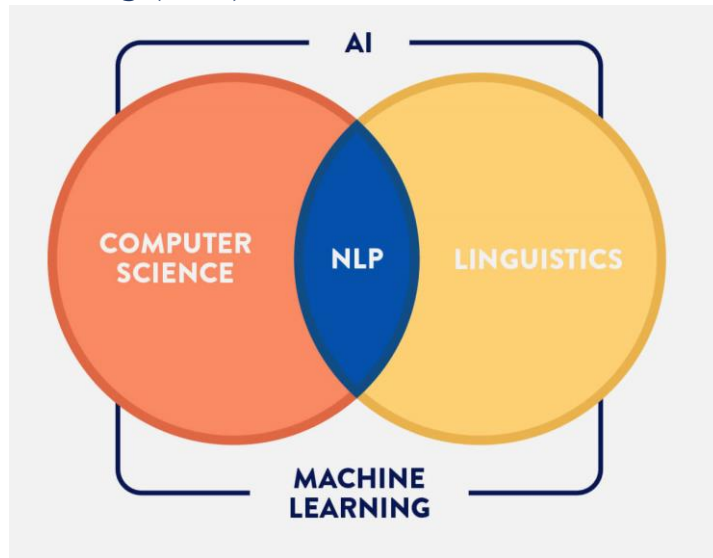
Overall, the architecture of a chatbot is designed to support its core functions of understanding user queries, generating appropriate responses, and managing conversation flow. By leveraging advanced technologies such as NLP, dialog management, and machine learning, chatbots can provide efficient and effective customer service while improving over time.





Chatbot

3.1 Natural Language Processing (NLP)



Natural Language Processing (NLP) is a branch of artificial intelligence (AI) that focuses on enabling computers to understand and interpret human language. It involves the development of algorithms and computational models that can analyze and process large volumes of human language data, including written text, spoken language, and even sign language.

NLP is an essential component of chatbots, as it enables the chatbot to understand and interpret user queries in a natural and conversational way. Some of the key components of NLP include:

Text Preprocessing: This involves cleaning and standardizing the raw text data, such as removing punctuation and converting all text to lowercase.

Tokenization: This involves breaking down the text into smaller units, such as words or phrases, to facilitate further analysis.

Part-of-Speech (POS) Tagging: This involves assigning a grammatical label to each word in the text, such as noun, verb, adjective, or adverb.

Named Entity Recognition (NER): This involves identifying and categorizing named entities in the text, such as people, organizations, locations, and dates.

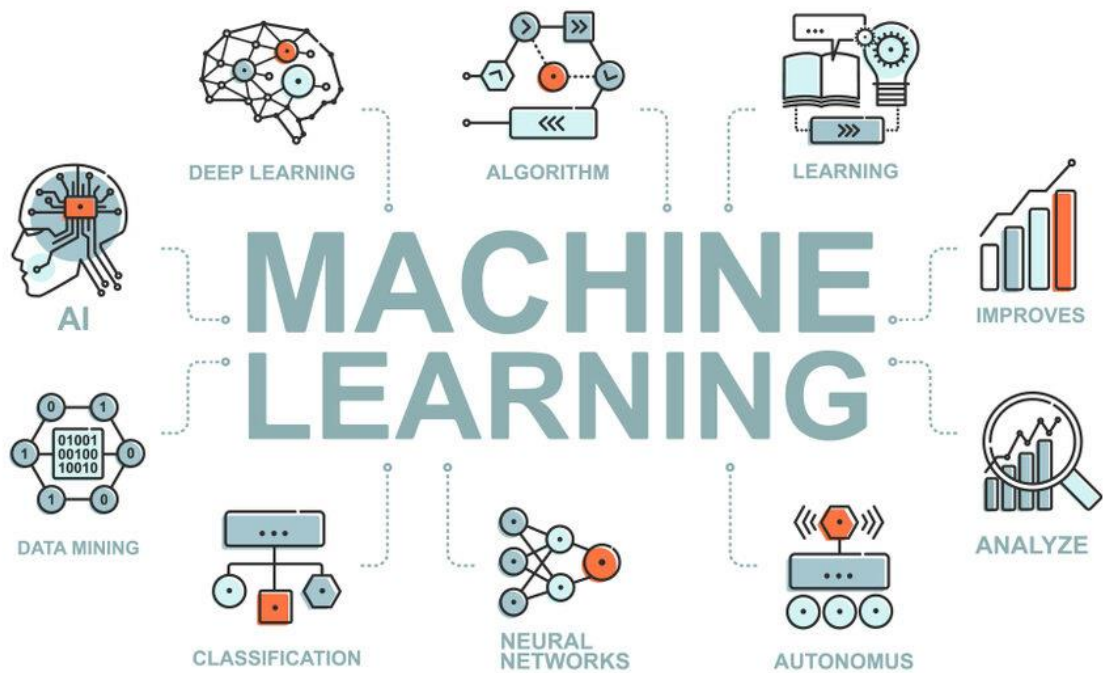
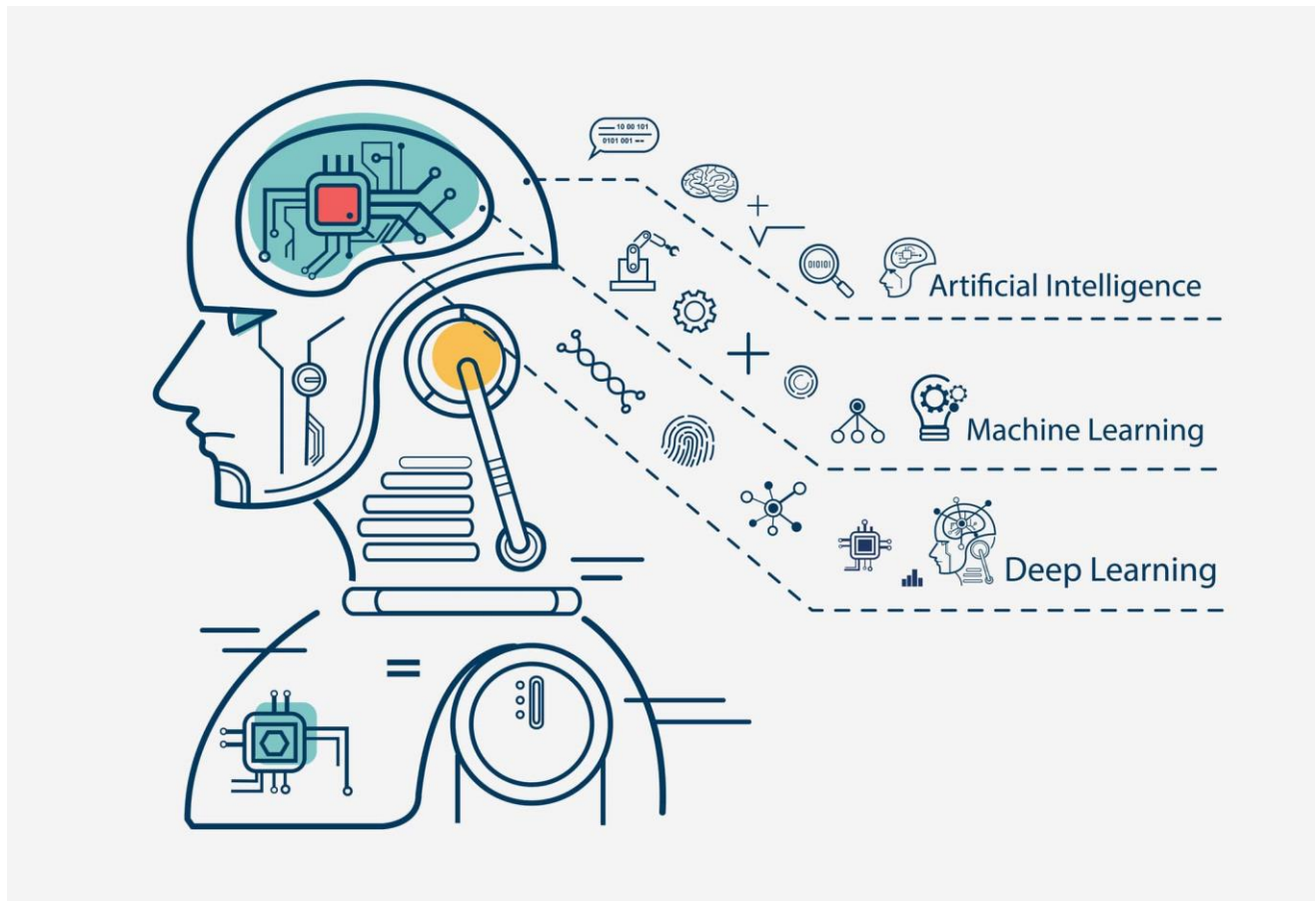
Sentiment Analysis: This involves analyzing the emotional tone of the text, such as positive, negative, or neutral.

Syntax Analysis: This involves analyzing the grammatical structure of the text, such as identifying subject-verb-object relationships.

Overall, NLP enables chatbots to understand and interpret user queries in a more sophisticated and nuanced way, allowing for more natural and conversational interactions. By leveraging advanced NLP algorithms and models, chatbots can provide efficient and effective customer service while improving over time.

Chatbot

3.2 Machine Learning





Chatbot

Machine learning is a subset of artificial intelligence (AI) that involves the development of algorithms and models that can learn from data and make predictions or decisions based on that data. Machine learning enables computers to automatically improve their performance over time by learning from experience, without being explicitly programmed.

In the context of chatbots, machine learning is an essential component that enables the chatbot to learn and improve over time based on user interactions. Some of the key applications of machine learning in chatbots include:

Intent Recognition: Machine learning algorithms can analyze user queries and determine the user's intent based on previous interactions. This enables the chatbot to provide more accurate and relevant responses.

Response Generation: Machine learning algorithms can generate responses based on patterns and insights derived from previous interactions. This enables the chatbot to provide more personalized and natural responses.

Contextual Understanding: Machine learning algorithms can analyze the context of the conversation, including previous interactions and user preferences, to provide more personalized and relevant responses.

Sentiment Analysis: Machine learning algorithms can analyze the emotional tone of user queries and responses, allowing the chatbot to provide more empathetic and appropriate responses.

Performance Optimization: Machine learning algorithms can analyze user interactions and performance metrics to optimize the chatbot's performance over time.

Overall, machine learning is a critical component of chatbot technology, enabling chatbots to learn from user interactions and provide increasingly sophisticated and personalized responses. By leveraging advanced machine learning algorithms and models, chatbots can provide efficient and effective customer service while continuously improving over time.



Chatbot

4. History of chatbots – chatbot development over the course of time

The first chatbot ever was developed by MIT professor Joseph Weizenbaum in the 1960s. It was called ELIZA. You'll read more about ELIZA and other popular chatbots that were developed in the second half of the 20th century later on.

In the year 2009, a company called **WeChat** in China created a more advanced Chatbot. Since its launch, WeChat has conquered the hearts of many users who demonstrate an unwavering loyalty to it. It is a highly thriving social media platform.

Through its platform, it has made it easy to create very simple chatbots. It has grown to be an example of the most favored ways for marketers and employers to reduce the work they do as they interact with customers online.

Though it has implications and is less performant than today's messaging apps such as Facebook Messenger, Slack, and Telegram, it doesn't mean that you cannot construct a very smart bot on WeChat. **Chumen Wenwen** Company, founded in 2012 by a former Google employee, has built a very sophisticated bot running on WeChat.

Early in 2016, we saw the intro of the first wave of artificial data technology in the design of chatbots. Social media platforms like Facebook enabled developers to build a chatbot for their trademark or service so that customers could carry out some of their daily actions from inside their messaging platform.

The introduction of chatbots into a community has brought us to the time of the conversational interface. It's an interface that soon won't demand a screen or a mouse. The interface will be entirely conversational, and those communications will be indistinguishable from the conversations that we have with our friends and relatives.

To fully explain the massiveness of this soon-to-be reality, we'd have to go back to the earliest days of the computer, when the desire for artificial intelligence technology and a conversational interface first began.

Chatbot

4.1 History of chatbots – From ELIZA to ChatGPT



ELIZA

ELIZA was the very first chatbot as mentioned above. It was created by Joseph Weizenbaum in 1966 and it uses pattern matching and substitution methodology to simulate conversation.

The program was designed in a way that it mimics human conversation. The Chatbot ELIZA worked by passing the words that users entered into a computer and then pairing them to a list of possible scripted responses. It uses a script that simulated a psychotherapist. The script proved to be a significant impact on natural language processing and unnatural intelligence, with copies and variants protruding up at academies around the country.

However, Weizenbaum was troubled by the reaction of users. He intended ELIZA to be a mere caricature of human conversation, yet suddenly users were confiding their most profound thoughts in ELIZA. Experts were declaring that chatbots would be indistinguishable from humans within a few number of years.



Chatbot

Welcome to

```
EEEEEE LL      IIII  ZZZZZZ  AAAAA
EE      LL      II    ZZ      AA  AA
EEEEEE LL      II    ZZZ      AAAAAA
EE      LL      II    ZZ      AA  AA
EEEEEE LLLLLL IIII  ZZZZZZ  AA  AA
```

Eliza is a mock Rogerian psychotherapist.

The original program was described by Joseph Weizenbaum in 1966.

This implementation by Norbert Landsteiner 2005.

ELIZA: Is something troubling you ?

YOU: Men are all alike.

ELIZA: What is the connection, do you suppose ?

YOU: They're always bugging us about something or other.

ELIZA: Can you think of a specific example ?

YOU: Well, my boyfriend made me come here.

ELIZA: Is it important to you that your boyfriend made you come here ?

YOU: He says I'm depressed much of the time.

ELIZA: I am sorry to hear that you are depressed.

YOU: It's true. I am unhappy.

ELIZA: Can you explain what made you unhappy ?

YOU:

Weizenbaum rejected the notion that machines could replace human intellect. He argued instead that such devices were just tools, and extensions of the human mind. He further stressed that computers' understanding of language was entirely dependent on the context in which they were used. Furthermore, Weizenbaum argued that a more general computer understanding of human language was not possible.

In the decades that followed, chatbot makers have built upon Weizenbaum's model to strive for more human-like interactions. Passing the Turing test has grown to a common goal, which tests new bots' conversational talents against a board of human judges. The hardest thing in the Turing test issue is that there's no limit on what people can discuss.

PARRY

PARRY was constructed by American psychiatrist Kenneth Colby in 1972. The program imitated a patient with schizophrenia. It attempts to simulate the disease. It is a natural language program that resembles the thinking of an individual.

PARRY works via a complicated system of assumptions, attributions, and "emotional responses" triggered by changing weights assigned to verbal inputs. To validate the work, PARRY was tested using a variation of the Turing test. It was in the early seventies when human interrogators, interacting with the program via a remote keyboard, were weak with more than random accuracy to distinguish PARRY from an original unreasonable individual.

Fifty years ago Kenneth Mark Colby was the only psychiatrist thinking about how computers could contribute to the understanding of mental illness. He thus began the project „Overcoming Depression” that lasted until his death in 2001.



Chatbot

Jabberwacky

The chatbot was created by developer Rollo Carpenter in 1988. It aimed to simulate a natural human conversation in an entertaining way.

Jabberwacky has led to other technological growth. Some individuals use it for academic research purposes through its webpage since its origin.

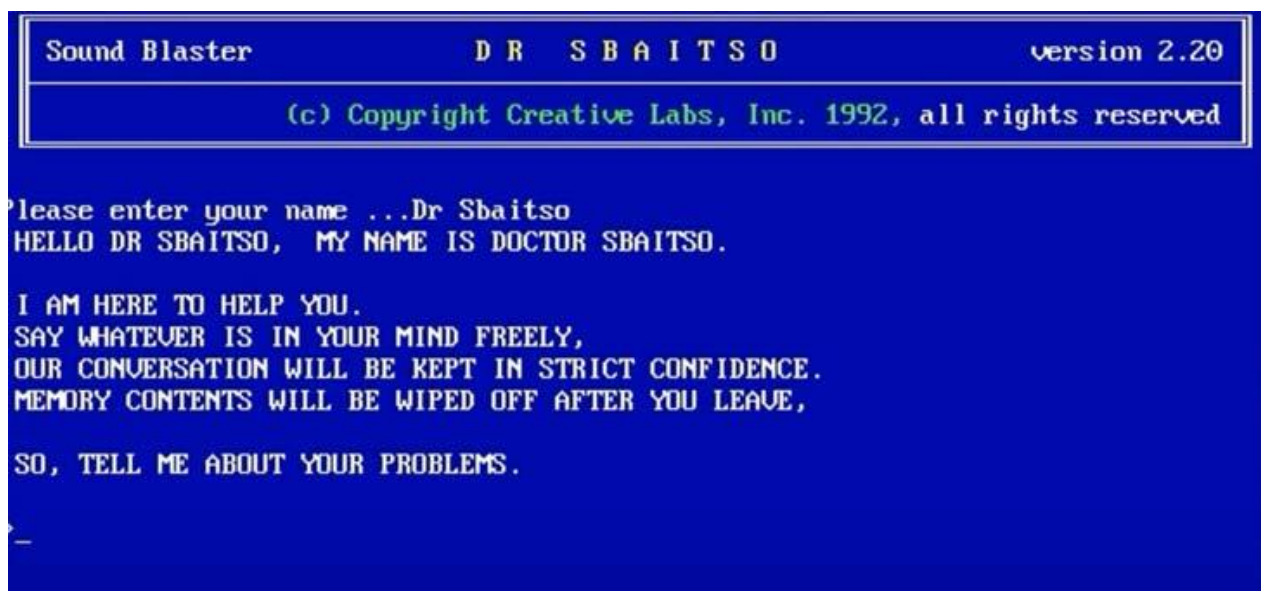
The chatbot is considered to use an AI technique called “contextual pattern matching.”

Dr. Sbaitso

Dr. Sbaitso is a chatbot created by Creative Labs for MS-Dos in 1992.

It is one of the earliest efforts of incorporating A.I. into a chatbot and is recognized for its full voice operated chat program.

The program would converse with the user as if it was a psychologist. Most of its responses were along the lines of “Why do you feel that way?” rather than any sort of complicated interaction.



A.L.I.C.E. (Artificial Linguistic Internet Computer Entity)



A.L.I.C.E. is a universal language processing chatbot that uses heuristic pattern matching to carry conversations. In 1995, Richard Wallace pioneered the construction of ALICE. It was formerly known as Alicebot because it was first to run on a computer by the name of Alice.



Chatbot

The program works with the XML schema known as artificial intelligence markup language (AIML), which helps specify conversation rules. In 1998, the program was edited in Java, and in 2001 Wallace printed an AIML specification. From there, other developers drafted free and open sources of ALICE in different programming languages and a variety of foreign languages.

The program simulates chatting with a real person over the Internet. Alice is a young-looking woman in human years and tells a user her age, hobbies and other fascinating facts, as well as answering to the user's dialog.

A. L. I. C. E. Artificial Intelligence Foundation
Promoting the development and adoption of **ALICE** and **AIML Free Software**

[A.L.I.C.E. Silver Edition](#) [DAVE E.S.L. Bot](#) [C.L.A.U.D.I.O Personality Test](#) [GET SITEPAL AVATARS](#)

Get your own free chat bot: Be Your Own Botmaster, 2nd ed.

Click here to chat with Talking Animated Fake Captain Kirk

A.L.I.C.E. and judge
You said: how are you
A.L.I.C.E.: My logic and cognitive functions are normal.

You say: Say

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Conversation Log:
judge: hi
ALICE: Hi there!
judge: how are you
ALICE: My logic and cognitive functions are normal.

Get ALICE Showcase Edition with talking Sitepal avatar.
CallMom ALICE 2.0 Android app
English Tutor app
Dave ESL Bot

Just remember, this was released in 1995, so don't expect too much in terms of user interface and design.

SmarterChild



The **SmarterChild** was in many ways a precursor of Siri and was developed in 2001.

The chatbot was available on AOL IM and MSN Messenger with the strength to carry out fun conversations with quick data access to other services.

It suits Microsoft also built its own SmarterChild, years later after most people stopped using AIM which targeted 18- to 24-year-olds in the U.S. the account suites particular conversation

Chatbot

Siri



Siri was formed by Apple for iOS in 2010; it is an intelligent personal assistant and learning navigator that uses a natural language UI. It paved the system for all AI bots and PAs after that.

A patent application by the United States Patent and Trademark Office details a new Apple service where users could make inquiries and conversation with Siri through Messages. The new patent is similar to a published late last year, but now includes deeper integration with audio, video, and image files.

Similar to other texting and Facebook Messenger Apple's patent describes a Siri that could perform current duties without the user having to chat aloud. That could be helpful in several public spheres.

They could reply to a text, audio, images, and video when transferred to it by the user. Apple said this would result in more fruitful interactive experience among a consumer and a digital assistant.

The patent provides a few examples of a conversation held between Siri and a user in Messages, with the user asking questions.

Google Now/Google Assistant



Google Now was launched at Google Inc in 2012. It answers questions, performs actions through requests made to a set of web services and makes recommendations.

It was part of a package of updates and UI modifications for mobile search, which included a female-voiced portable assistant to compete with Apple's Siri.

Google Now was initially a way to get contextually appropriate information based on location and time of the day. It evolved to become much more complicated and elaborate, with a broad range of content categories delivered on cards.

Sometimes it refers to us as predictive search. Currently, it's built for use in smartphones and has been upgraded to accommodate several features.

Google Now was replaced by Google Assistant in 2017. Today, the assistant is part of a more aggressive Google search growth strategy. The idea is simple, Google wants to provide information in an easy-to-read format before you even know you need it.

Chatbot

Cortana



Cortana was first demonstrated at Microsoft's Build 2014 developer conference, and it became directly integrated into both Windows phone devices and Windows 10 PCs.

This program uses voice recognition and relevant algorithms to get and respond to voice commands.

For someone to get started, they must type a question in the search box, or select the microphone and talk to Cortana. If a person is not very sure of what to say, they will see suggestions on the lock screen, as well as in Cortana home by selecting the search box on the taskbar.

Cortana can perform tasks like reminders based on time, places, or people, send emails and texts, create and manage lists, chit-chat, and play games, find facts, files, locations, and info among others.

Alexa



amazon echo dot



Alexa is an intelligent personal assistant developed by Amazon. It was introduced in 2014 and is now built in to devices such as the Amazon Echo, the Echo Dot, the Echo Show and more. There is also an Alexa app and more devices from third-party manufacturers that have Alexa built in to them.

All you have to do is say “Alexa, play some music” or “Alexa, find me an Italian restaurant” and she will help you out.

Using nothing but the sound of your voice, you can search the Web, play music, create to-do or shopping lists, set alarms, stream podcasts, play audiobooks, get news or weather reports, control your smart-home products and more.

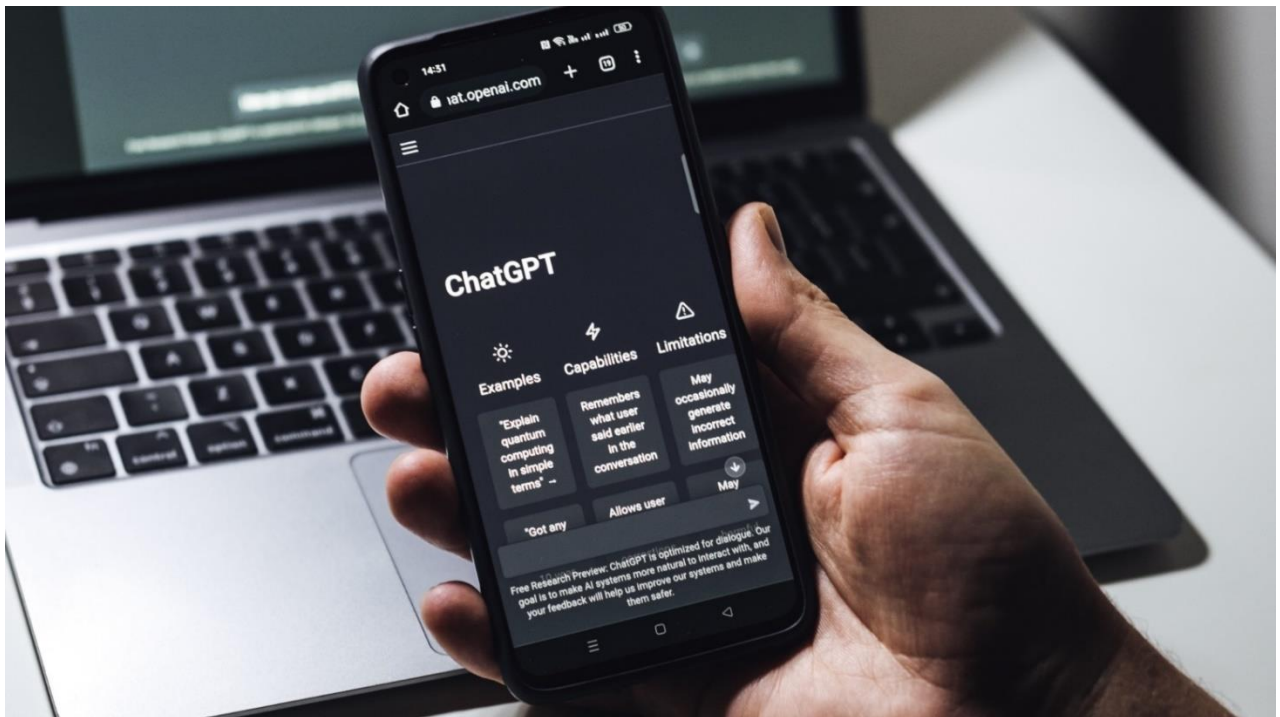
To add to the capabilities of any Alexa-enabled device, Amazon allows developers to build and publish skills for Alexa using the Alexa Skills Kit (ASK). You can download skills for free with the Alexa app.

Chatbot

ChatGPT

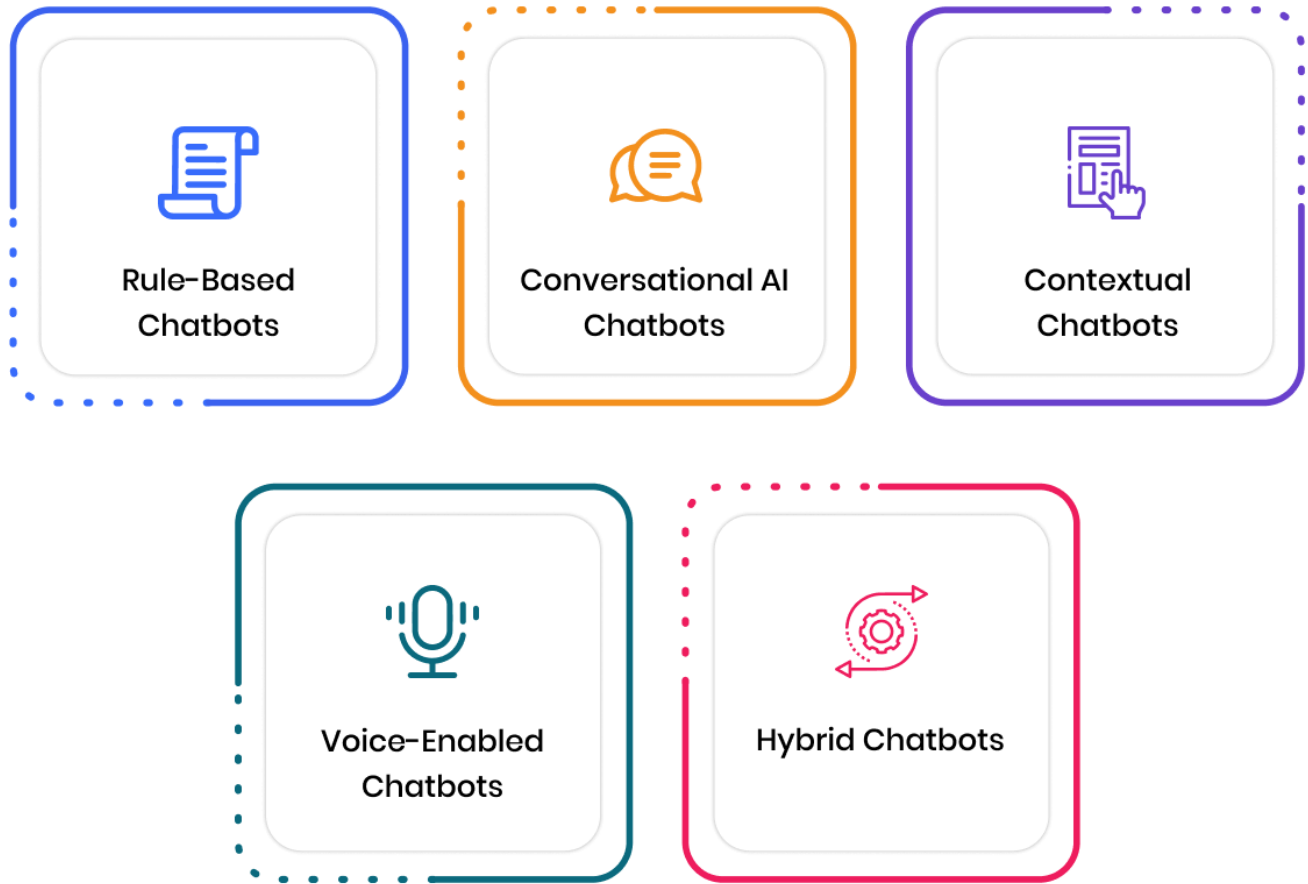
ChatGPT is a large language model trained by OpenAI. It was founded by the OpenAI team in 2021. It is designed to assist users in generating human-like text based on given input. ChatGPT can be used for a variety of tasks, including conversation generation and language translation.

The model is trained on a massive amount of data, allowing it to generate text that is often difficult to distinguish from text written by a human. ChatGPT has been praised for its ability to generate natural-sounding text and its potential applications in a variety of fields. By the way – this abstract was created by ChatGPT.





5. Types of Chatbots



There are several types of chatbots, each with their own unique characteristics and applications. Some of the most common types of chatbots include:

Rule-based chatbots: Rule-based chatbots are the most basic type of chatbot, relying on a predetermined set of rules and responses. They are typically used for simple and repetitive tasks, such as answering frequently asked questions or providing basic customer support.

AI-powered chatbots: AI-powered chatbots leverage advanced technologies such as natural language processing (NLP) and machine learning to provide more sophisticated and personalized responses. They can understand and interpret user queries in a more nuanced way, allowing for more natural and conversational interactions.

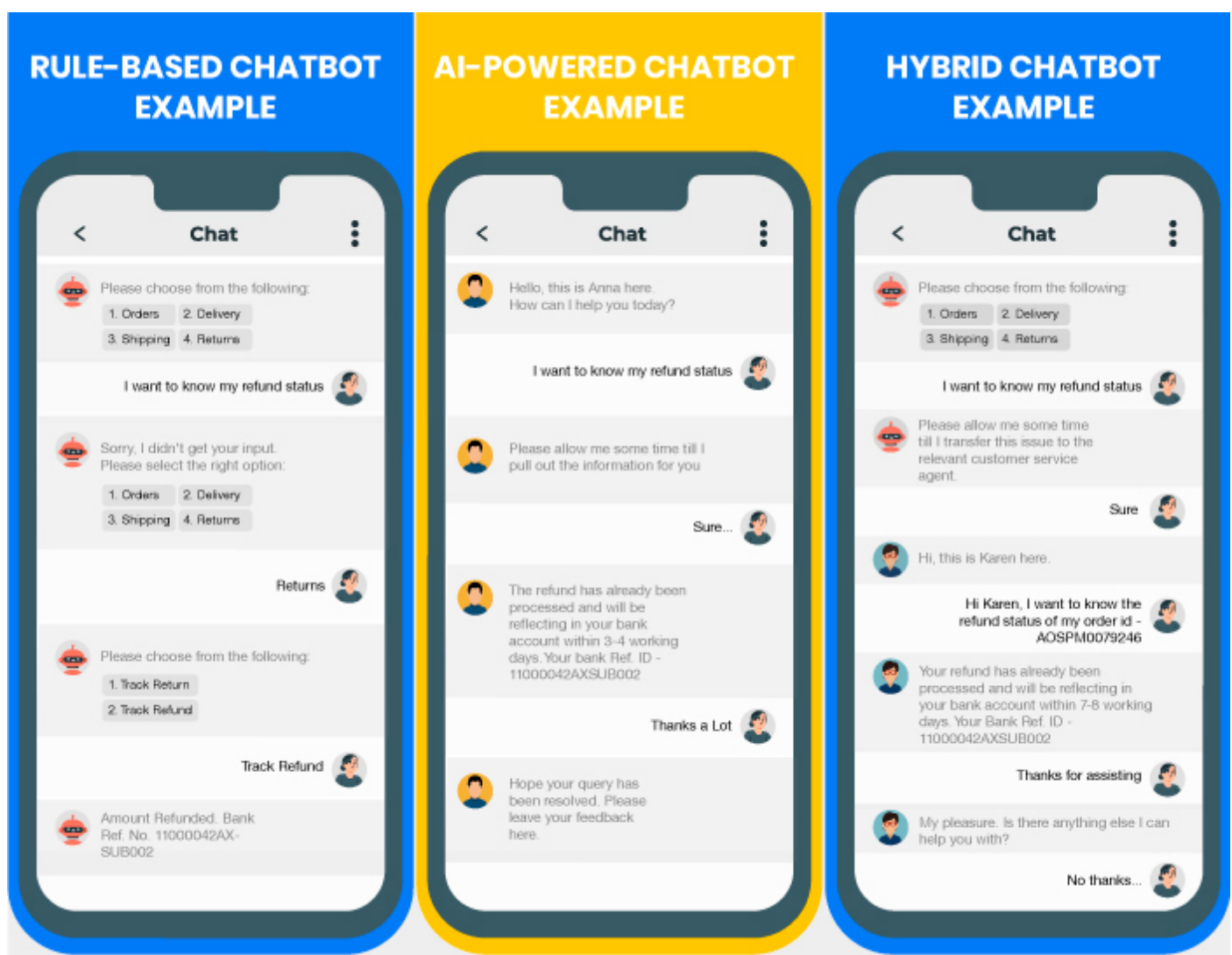
Voice-activated chatbots: Voice-activated chatbots are designed to interact with users via spoken language, typically using voice assistants such as Amazon Alexa or Google Assistant. They are commonly used for tasks such as setting reminders, playing music, or controlling smart home devices.

Chatbot

Social media chatbots: Social media chatbots are designed to interact with users on social media platforms such as Facebook, Twitter, or Instagram. They can be used for a variety of tasks, such as providing customer support, running marketing campaigns, or engaging with users through quizzes or games.

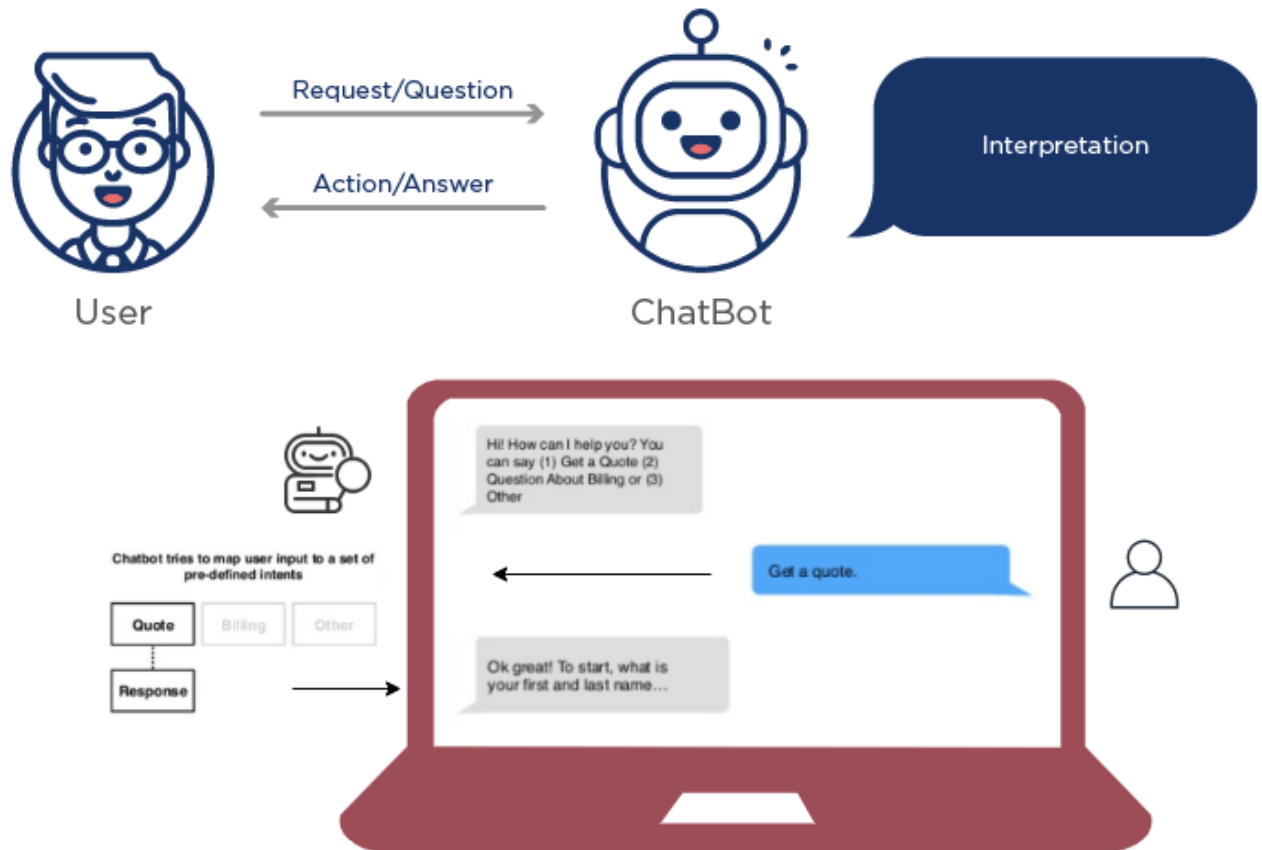
Virtual assistants: Virtual assistants are advanced chatbots that can perform a wide range of tasks, such as scheduling appointments, making reservations, or providing weather forecasts. They typically integrate with other software applications and services to provide a seamless user experience.

Overall, the type of chatbot used will depend on the specific needs and applications of the business or organization. By leveraging the right type of chatbot, businesses can provide efficient and effective customer service while improving customer engagement and satisfaction.



Chatbot

5.1 Rule-based chatbots



Rule-based chatbots are the simplest type of chatbot, relying on a set of predefined rules to generate responses to user queries. The rules are typically based on a decision tree or flowchart that maps out all possible user inputs and corresponding responses.

The rules in a rule-based chatbot are based on a set of predefined keywords or phrases that the chatbot is programmed to recognize. When a user inputs a query, the chatbot matches the query to the predefined rules and generates a response based on the corresponding rule.

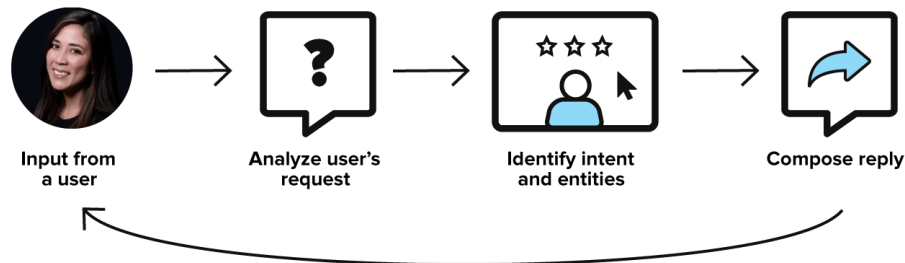
One of the main advantages of rule-based chatbots is their simplicity and ease of implementation. They are relatively easy to build and require minimal training data or machine learning algorithms. They are also effective for simple and repetitive tasks, such as answering frequently asked questions or providing basic customer support.

However, rule-based chatbots have some limitations. They can only respond to queries that match the predefined rules, and cannot generate responses that go beyond the scope of the predefined rules. They also have limited flexibility and may struggle with more complex or ambiguous queries. As a result, rule-based chatbots are best suited for simple and straightforward tasks, and may not be suitable for more complex or sophisticated applications.

Chatbot

5.2 AI-powered chatbots

HOW AN A.I. CHATBOT WORKS



AI-powered chatbots, also known as intelligent chatbots, leverage advanced technologies such as natural language processing (NLP) and machine learning to provide more sophisticated and personalized responses. They can understand and interpret user queries in a more nuanced way, allowing for more natural and conversational interactions.

AI-powered chatbots can learn from previous interactions to improve their performance over time. They use machine learning algorithms to analyze user queries and identify patterns and insights that can be used to generate more accurate and relevant responses.

One of the main advantages of AI-powered chatbots is their ability to understand and interpret natural language. They can recognize synonyms and variations of words, and can identify the underlying intent behind user queries. This allows for more natural and conversational interactions, which can improve user engagement and satisfaction.

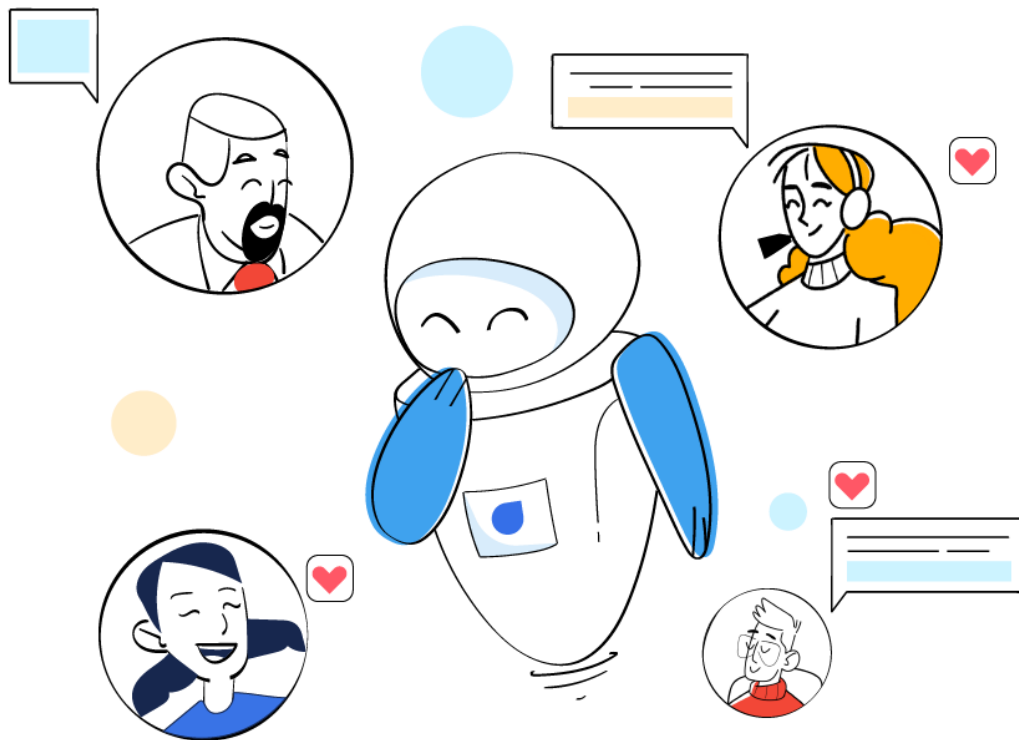
AI-powered chatbots can also handle more complex queries and tasks than rule-based chatbots. They can provide personalized recommendations, make predictions based on user data, and perform more sophisticated tasks such as booking appointments or making reservations.

However, AI-powered chatbots require more advanced technology and expertise to build and maintain. They typically require large amounts of training data and complex machine learning algorithms to perform effectively. Additionally, they may require ongoing monitoring and optimization to ensure that they are providing accurate and relevant responses.

Overall, AI-powered chatbots are a powerful tool for businesses looking to provide efficient and effective customer service while improving user engagement and satisfaction. By leveraging advanced technologies such as NLP and machine learning, AI-powered chatbots can provide personalized and natural interactions that improve the overall user experience.

Chatbot

5.3 Hybrid chatbots



Hybrid chatbots are chatbots that combine the features of rule-based and AI-powered chatbots. They use a combination of predefined rules and machine learning algorithms to provide more accurate and relevant responses to user queries.

Hybrid chatbots typically start with a set of predefined rules that are used to generate responses to common queries. However, as users interact with the chatbot, it collects data on their preferences and behavior. This data is then used to train machine learning algorithms that can identify patterns and insights that can be used to generate more personalized and accurate responses.

One of the main advantages of hybrid chatbots is their ability to provide a balance between flexibility and accuracy. They can handle a wide range of queries and tasks, while also providing personalized and relevant responses based on user data.

Hybrid chatbots are also more adaptable than rule-based chatbots, as they can learn and evolve over time. As they collect more data and refine their algorithms, they can become more accurate and effective at handling complex user queries and tasks.



Chatbot

However, hybrid chatbots can be more complex to build and maintain than rule-based chatbots. They require expertise in both rule-based and AI-powered chatbot development, as well as ongoing monitoring and optimization to ensure that they are providing accurate and relevant responses.

Overall, hybrid chatbots are a powerful tool for businesses looking to provide efficient and effective customer service while improving user engagement and satisfaction. By combining the benefits of rule-based and AI-powered chatbots, hybrid chatbots can provide a more personalized and natural user experience that improves overall customer satisfaction.

6. Advantages of Chatbots

Chatbots Benefits in Customer Service



Enhanced customer experience



Easier support for multiple languages



Non-intrusive customer profiling



Advanced personalization



Round-the-clock support



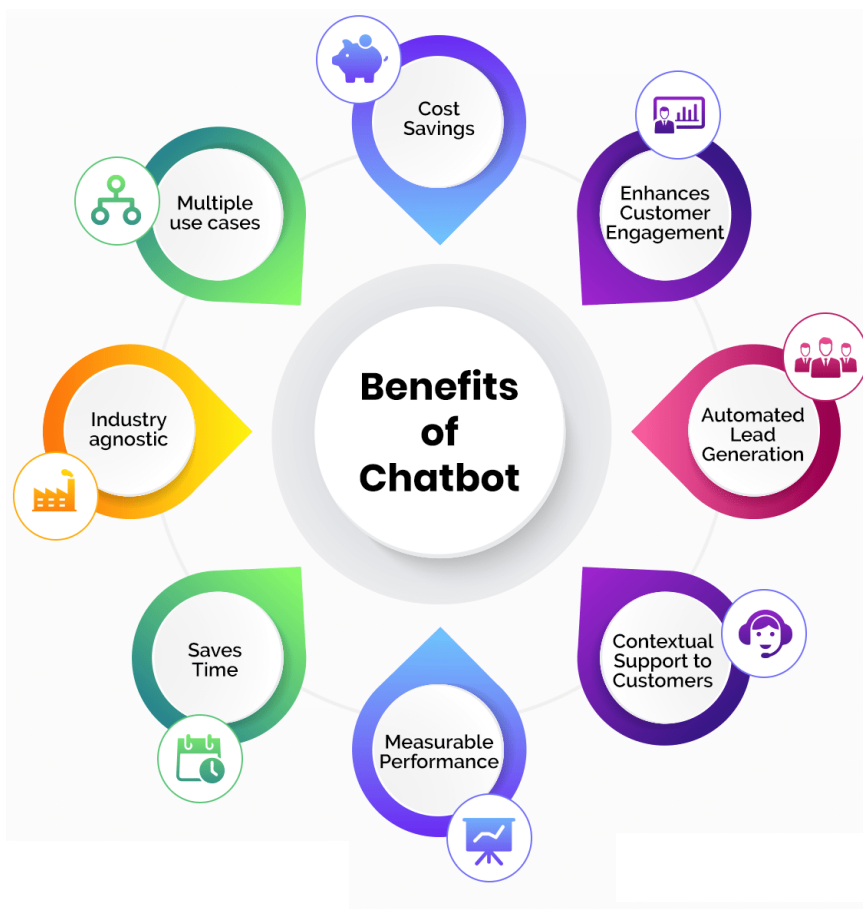
Improved brand recognition



New revenue channels



Automation of processes



Chatbot

There are several advantages of chatbots that make them a valuable tool for businesses:

Available 24/7: Chatbots can operate around the clock, providing customers with immediate assistance and support at any time of the day or night. This can improve customer satisfaction and loyalty, as customers can get help whenever they need it.

Scalability: Chatbots can handle multiple queries and interactions simultaneously, making them highly scalable. This can be particularly useful for businesses that experience high volumes of customer queries and support requests.

Cost-effective: Chatbots can be more cost-effective than human customer service representatives, as they can handle a large volume of interactions without the need for additional staff. This can help businesses reduce costs while still providing effective customer service.

Consistency: Chatbots can provide consistent responses to customer queries, ensuring that customers receive accurate and reliable information every time. This can help build trust and credibility with customers.

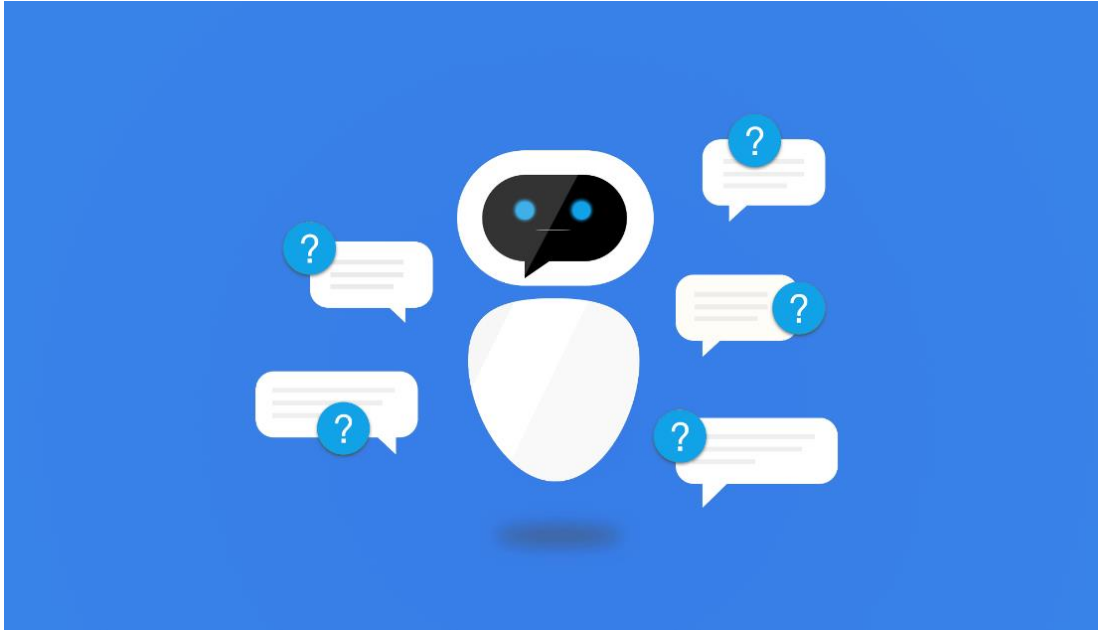
Personalization: AI-powered chatbots can use machine learning algorithms to analyze customer data and provide personalized recommendations and responses. This can improve the overall user experience and increase customer satisfaction.

Efficiency: Chatbots can handle routine and repetitive tasks, such as answering frequently asked questions and processing orders, freeing up human customer service representatives to focus on more complex tasks and interactions.

Overall, chatbots can provide businesses with a range of benefits, including improved customer satisfaction, cost savings, and increased efficiency. By leveraging the latest technologies, businesses can create chatbots that provide personalized and effective support to customers, improving their overall experience with the brand.



7. *Limitations of Chatbots*



Limited Understanding: Chatbots can struggle with understanding and interpreting complex queries or questions. They may not be able to understand sarcasm, irony, or colloquial language, which can lead to inaccurate or irrelevant responses.

Lack of Empathy: Chatbots lack the emotional intelligence and empathy that human customer service representatives can provide. They may not be able to recognize and respond to emotional cues or tone of voice, which can negatively impact the user experience.

Technical Limitations: Chatbots are reliant on technology, which can be prone to errors and downtime. If the chatbot experiences technical difficulties, it may not be able to provide support to customers, which can lead to frustration and dissatisfaction.

Security Concerns: Chatbots may handle sensitive customer data, such as personal and financial information. If not properly secured, this data can be vulnerable to hacking and data breaches.

Cost of Development: Developing and maintaining a chatbot can be expensive, particularly for businesses that require more sophisticated chatbots with AI and machine learning capabilities.

Need for Constant Monitoring: Chatbots require ongoing monitoring and maintenance to ensure that they are providing accurate and relevant responses. Without regular updates and optimization, chatbots can become outdated and ineffective.

Overall, while chatbots can be a valuable tool for businesses, it is important to consider their limitations and to use them in conjunction with human customer service representatives to provide the best possible customer experience.



Chatbot

8. *Designing a Chatbot*

Designing a chatbot can be a complex task, but here are some general steps you can follow:

Define the purpose and goals of your chatbot: Determine what the chatbot will be used for, and what goals you want to achieve with it. This will guide the design and development of the chatbot.

Choose a platform: Decide on a platform to build your chatbot. There are many chatbot platforms available, including Botpress, Dialogflow, Microsoft Bot Framework, and many more.

Determine the conversation flow: Map out the conversation flow that your chatbot will follow. This involves defining the various stages of the conversation and what questions or prompts the chatbot will provide at each stage.

Write the dialogue: Write the dialogue that your chatbot will use. This should be conversational and easy to understand, and it should take into account the user's likely responses and questions.

Choose a chatbot personality: Choose a personality for your chatbot that matches the purpose and goals of the chatbot. This will help to make the chatbot more engaging and memorable.

Test and refine: Test your chatbot with real users and refine the design based on their feedback. This will help you to identify any issues or areas for improvement, and to make sure that the chatbot is meeting its goals.

Launch and monitor: Once your chatbot is ready, launch it and monitor its performance. This will help you to identify any issues or areas for improvement, and to make sure that the chatbot is meeting its goals.

- Defining use cases

Customer service: Chatbots can be used to provide automated customer service support, answering common questions and resolving issues.

E-commerce: Chatbots can assist with online shopping by helping customers find products, providing product recommendations, and processing orders.

Informational: Chatbots can provide information about a variety of topics, including news, weather, and local events.

Lead generation: Chatbots can be used to capture leads by engaging with visitors on a website or social media platform.

Personal finance: Chatbots can help users manage their finances by providing budgeting advice, tracking expenses, and answering financial questions.

Travel: Chatbots can assist with travel planning by providing information on flights, hotels, and local attractions.



Chatbot

Healthcare: Chatbots can provide medical advice and help patients manage their health by providing reminders for medication and appointments.

HR: Chatbots can assist with HR tasks such as recruiting, onboarding, and answering employee questions.

Education: Chatbots can be used for educational purposes, providing language learning or tutoring assistance.

Entertainment: Chatbots can be used for entertainment purposes, such as playing games or telling jokes.

- Choosing the right platform

Choosing the right platform for your chatbot will depend on several factors, such as your budget, technical expertise, and the features and functionality you require. Here are some of the most popular chatbot platforms and their key features:

Dialogflow: Dialogflow is a Google-owned platform that allows you to build conversational interfaces for websites, mobile applications, and messaging platforms. It has natural language processing capabilities and a drag-and-drop interface for building chatbots.

Botpress: Botpress is an open-source chatbot platform that allows you to build, deploy, and manage chatbots on various platforms. It has a visual flow editor for building chatbot conversations and integrates with various messaging platforms.

Microsoft Bot Framework: Microsoft Bot Framework is a platform for building intelligent bots for various messaging platforms, including Facebook Messenger, Skype, and Slack. It has natural language processing capabilities and a drag-and-drop interface for building chatbots.

Amazon Lex: Amazon Lex is a service that allows you to build conversational interfaces for your applications using voice and text. It uses automatic speech recognition and natural language understanding to build chatbots.

IBM Watson Assistant: IBM Watson Assistant is an AI-powered platform that allows you to build chatbots for websites, mobile apps, and messaging platforms. It has natural language processing capabilities and a drag-and-drop interface for building chatbots.

Chatfuel: Chatfuel is a chatbot platform that allows you to build chatbots for Facebook Messenger and Telegram. It has a visual chatbot builder and integrates with various APIs to provide advanced functionality.

When choosing a platform for your chatbot, consider the features and functionality you require, the technical expertise of your team, and the scalability and cost of the platform.



Chatbot

9. Chatbot Technologies

There are several technologies used in chatbots to enable natural language processing, machine learning, and conversation management. Here are some of the key technologies used in chatbots:

Natural Language Processing (NLP): NLP is a technology that enables chatbots to understand and interpret human language. NLP involves parsing and analyzing human language to identify intent and extract key information from user input.

Machine Learning (ML): Machine learning is a technology that allows chatbots to learn and improve over time. ML algorithms analyze data and make predictions based on patterns, enabling chatbots to provide better responses and recommendations.

Artificial Intelligence (AI): AI technologies, such as natural language processing and machine learning, enable chatbots to interact with users in a more human-like way. This helps to improve user engagement and satisfaction.

Application Programming Interfaces (APIs): APIs are used to connect chatbots with external services and data sources, such as weather APIs, e-commerce APIs, or payment APIs. This enables chatbots to provide more personalized and relevant information to users.

Dialog Management: Dialog management technologies help chatbots manage conversations with users. This involves keeping track of the conversation flow, understanding user intent, and providing appropriate responses.

Sentiment Analysis: Sentiment analysis technologies enable chatbots to understand the tone and sentiment of user input. This can help chatbots respond appropriately to user emotions and provide a better user experience.

Voice Recognition: Voice recognition technologies enable chatbots to understand and respond to spoken input. This is particularly useful for chatbots used in voice assistants or phone-based customer service applications.

Text-to-Speech: Text-to-speech technologies convert text into spoken output, enabling chatbots to provide audio responses to users.

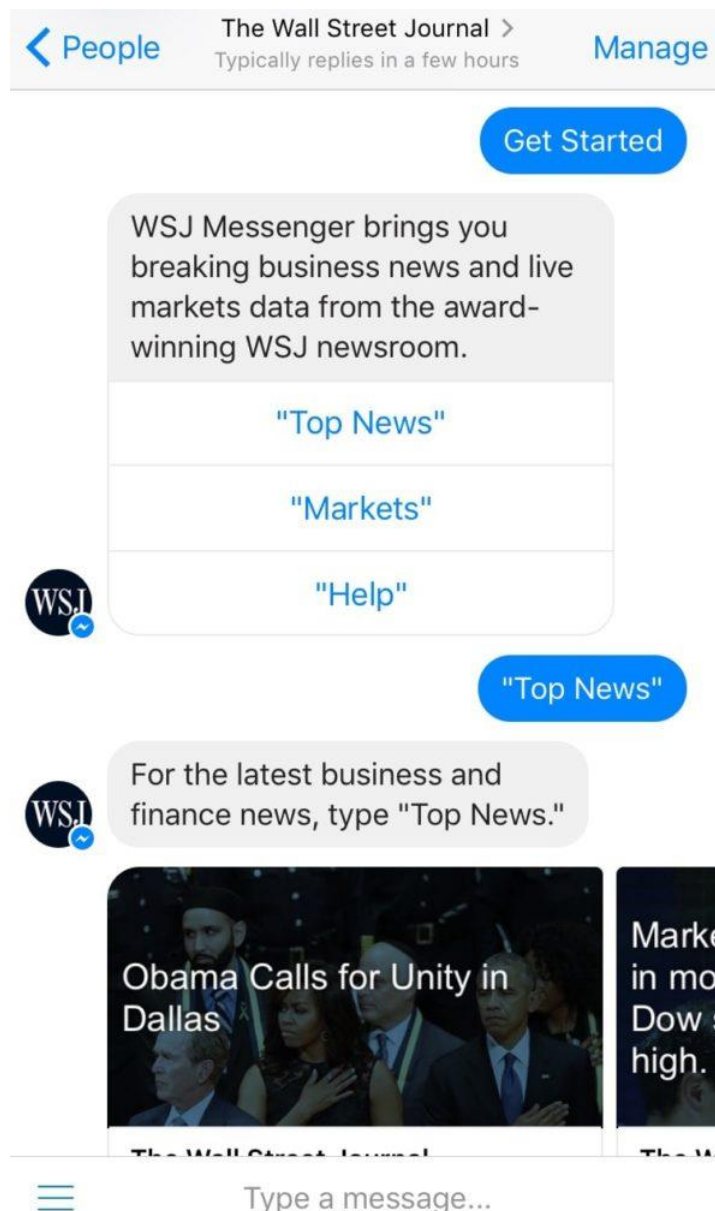
By leveraging these technologies, chatbots can provide a more natural and engaging conversation experience for users, leading to increased user satisfaction and improved business outcomes.

Chatbot

10. Applications where Chatbots are used

#1. Content delivery

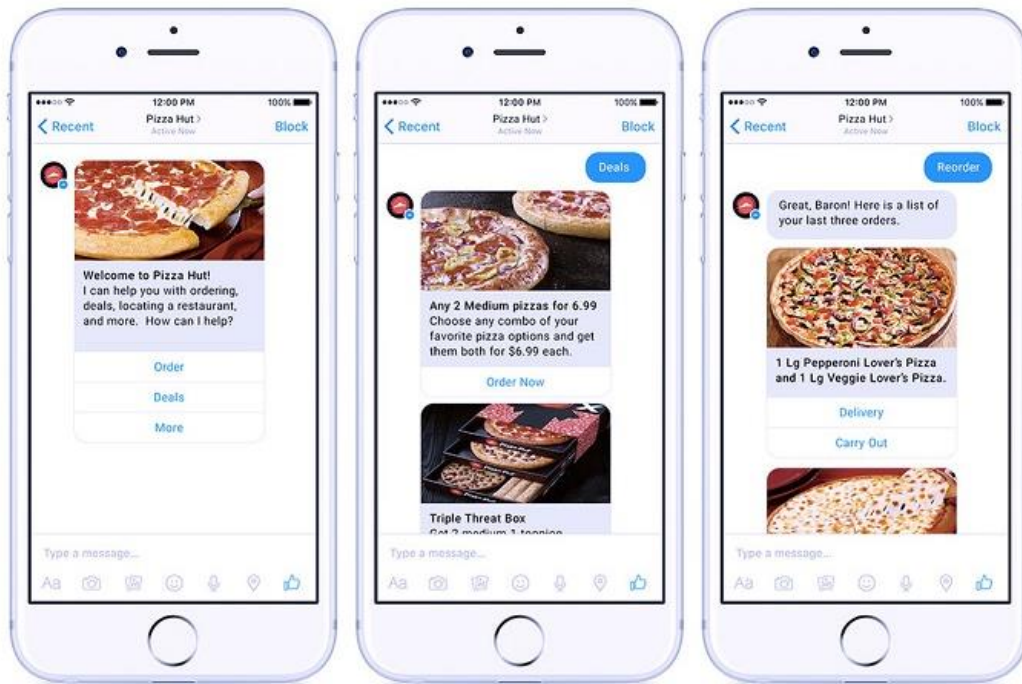
Media Publishers have realized that chatbots are a powerful way to engage with their audiences and monitor engagement to gain valuable insights on reader interests. Chat with the CNN and Wall Street Journal Chatbots on Facebook Messenger and receive the latest news directly in Messenger, without having to visit their websites.



A lot of publishers are also harnessing AI and machine learning technology within their chatbots to anticipate what content their consumers may be interested in. This is an interesting application of chatbot technology and its sure to lead to immediate and more personalized content than ever before.

Chatbot

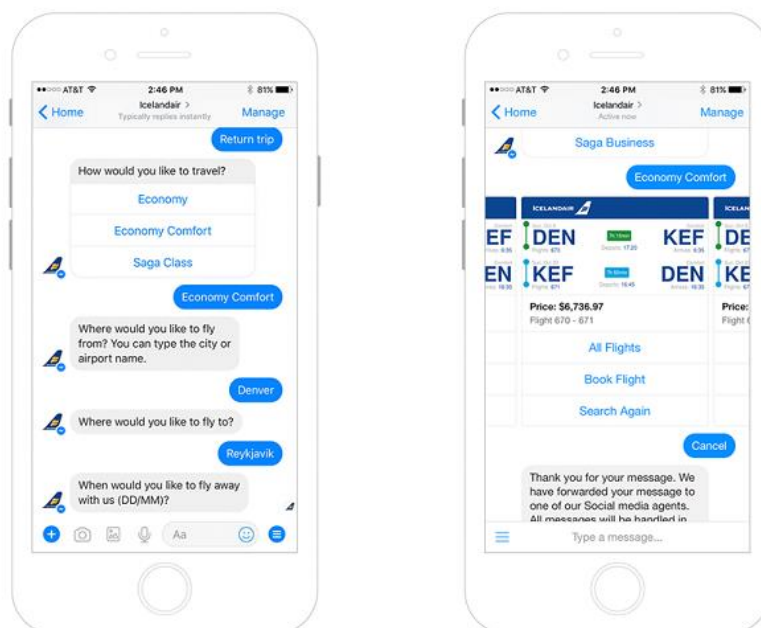
#2. Order Food



Various fast food giants like KFC and Pizza Hut have invested in Chatbots that enable customers to place their orders through conversations. Taco Bell went a step further to improve the conversational experience by giving their Chatbot named TacoBot some personality.

TacoBot cracks jokes, use emojis, answer trivia questions, and will even add a cup of water to an order if the customer mentions being hungover.

#3. Book Flights





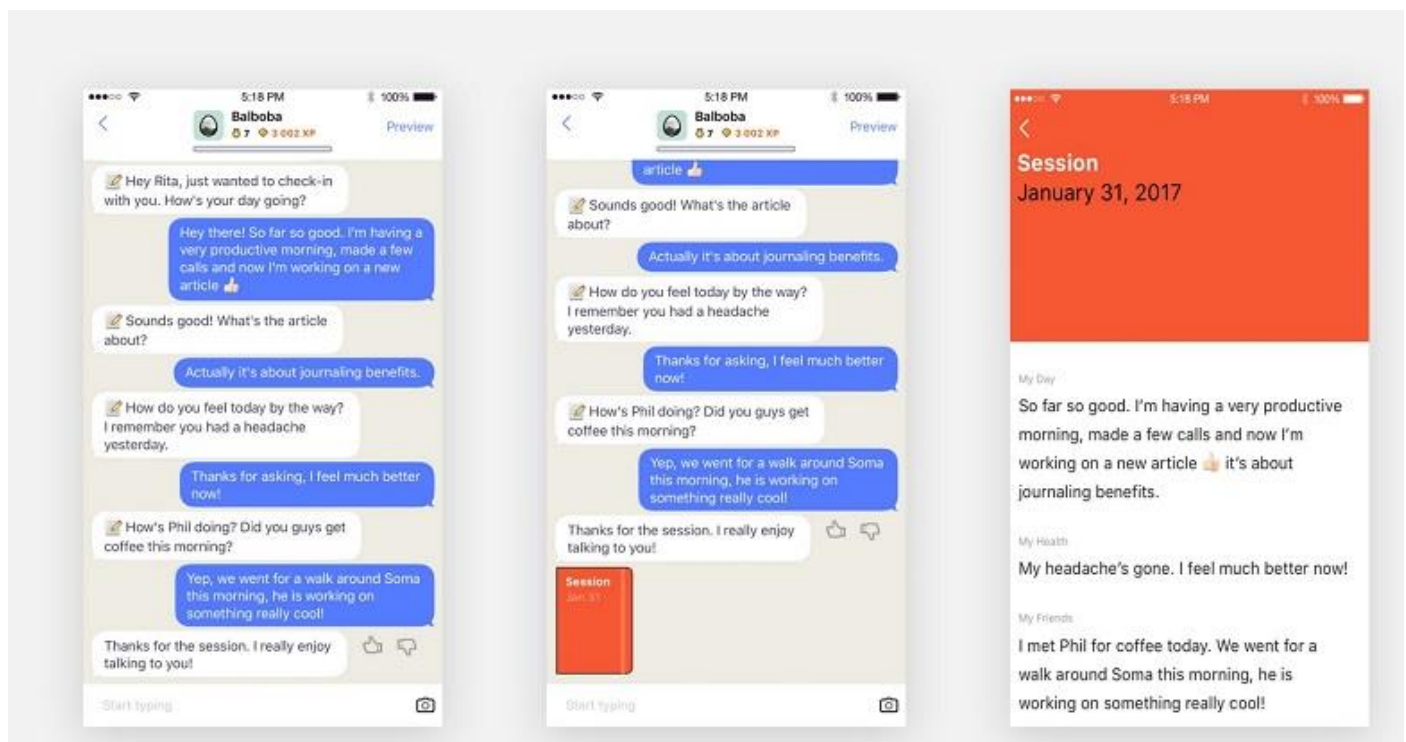
Chatbot

Airlines are also applying chatbots to their business operations, for instance, Icelandair's chatbot gives their customers the ability to search for and book flights in a text-based conversational manner.

Instead of drop-down menus, customers enter the information themselves. These features give customers more control over how the flight is booked and it also keeps the entire conversation in one thread so that the purchase information can be reviewed and called up with ease.

#4. Companionship

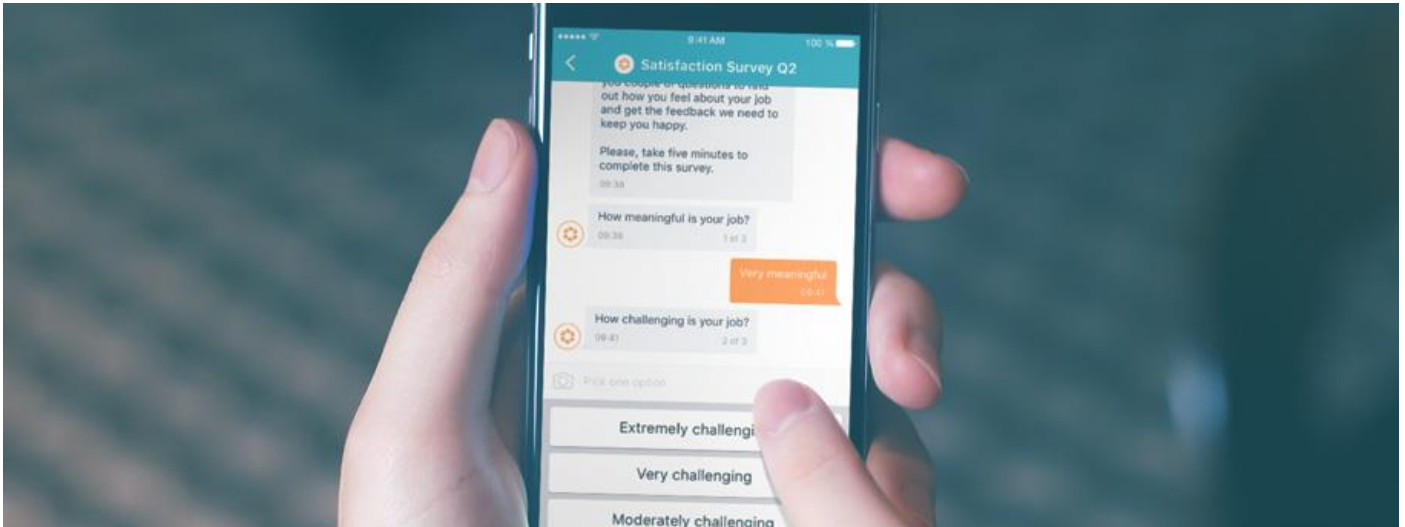
Russian technology company Endurance developed its companion chatbot for Senior People and Patients with Alzheimer's Disease. The primary function of the chatbot is to be a virtual companion – To speak with senior people on general topics like the weather, nature, hobbies, movies, music, news, etc. The chatbot asks questions, reacts to the answers, can speak on various topics, and share interesting news and facts from Google



Another companion bot worth mention is Replika. If you need a friend whenever you feel lonely, you can have a chat with [Replika](#). This bot is more like your A.I. friend whom you can tell everything – the bot learns from you, matches your personality and becomes your best friend. Replika can act as your safe space to share thoughts and feelings without fear of being judged. It is available for free on Google Play & Itunes.

Chatbot

#5. Market Research



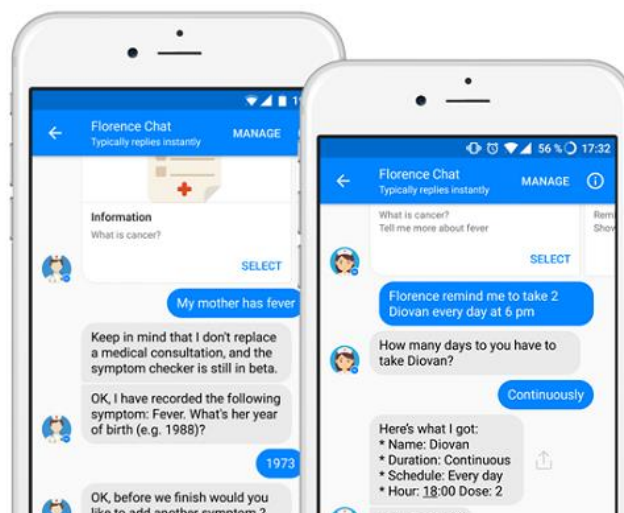
The way market researchers get in touch with respondents is changing as many respondents may prefer to be contacted on WhatsApp or the Facebook messenger. Instead of having a dedicated survey app, many research firms are developing Chatbots to have personalized, engaging conversational surveys with respondents to improve the experience and increase completion rates.

#6. Transportation

Uber in partnership with Facebook has enabled users to sign up for Uber and request a ride, without having to leave Messenger or download the Uber app. Ride status updates and ride receipts are delivered to a private conversation between the customers and Uber on Messenger, making it easy to track Uber rides and payment history.

#7. Health Care

Chatbots have also made their way into health care by easing the burden on medical professionals by facilitating faster medical diagnosis, answering health-related questions, booking appointments and lots more. A Chatbot like Super Izzy can track menstrual cycles, dates and fertile windows. The bot also asks menstruation-related questions and learns to give information about menstrual health and sexual issues.



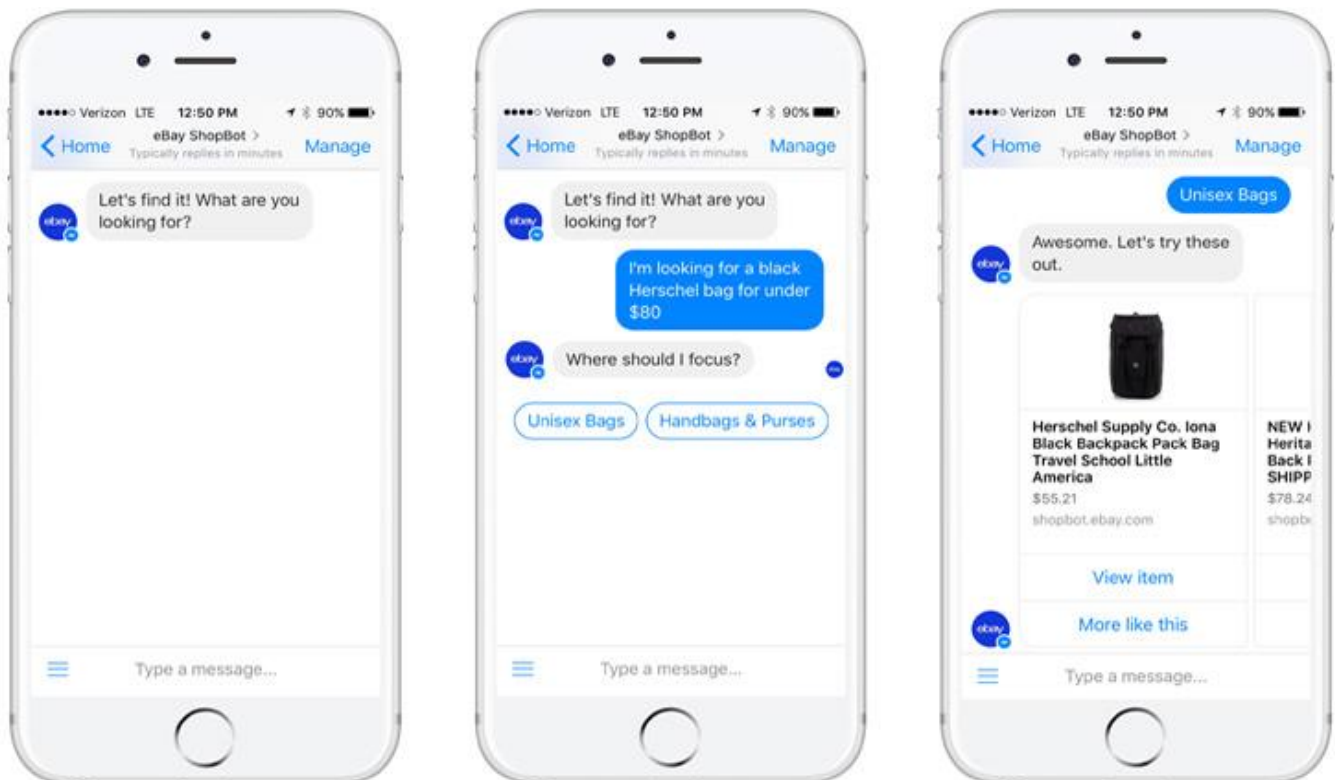


Chatbot

Another interesting Chatbot in health care is named Florence. This bot is more like a virtual nurse and has her office on Facebook Messenger. Florence can remind patients to take their pills, all they need to do is write the name of the medicine in chat, the number of times a day they must take it and at what time. Florence sends a message in chat every time the patient must take their meds

#8. E-commerce

The e-commerce industry is also improving shopping experience with Chatbots. Customers can now search and shop more conveniently with the help of chatbots. eBay has invested in Chatbot technology with their ShopBot, a virtual shopping assistant that helps customers find the items they want at the desired price range.



Customers only need to type in details of what they are looking (e.g. Telescopes under \$150), ShopBot will ask you all the relevant questions to find the best match. This a great example of how Chatbots can improve the shopping experience beyond what regular e-commerce sites can offer.

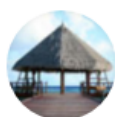
#9. Hospitality

Chatbots have been put to use in the hospitality industry in various ways. Radisson Blu Edwardian hotels use a virtual host called Edward to help guests with a variety of things from; online check-in, to handling customer inquiries, letting them know their rooms are ready and more.

Edward has his own personality, can automatically escalate complaints should a request be not followed up. The Chatbot is so good at his job that most guests don't know they are dealing with a ChatBot, and praised him with glowing reviews on TripAdvisor. The Edwardian Hotels team also leverages data from Edward to improve their services as they monitor common guest requests and trends.



Chatbot



Liolol
Birmingham, United Kingdom

58 22

Reviewed March 26, 2017

Great visit

Stayed here for a few nights. I checked in and was told my room, I thought someone would walk me to my room as I was expecting as I didn't really know where I was going, but wasn't guided... I love the way the whole hotel is access control - no one can use the lift or gym without a hotel key which is a first for any hotel view I've had. Great location, central to everything in London. The staff were very friendly. Ultra modern hotel! The rooms were extremely clean (I feel like the carpet could be replaced, looks dated), the bathroom was beautiful! **I love the virtual assistant, Edward! He was very friendly too and assisted me in all the ways he could!** The breakfast was exceptional - you had a breakfast buffet as well as your choice of cooked breakfast, I had smoked salmon, fruits with yoghurt (the fruits were so ripe and sweet!) and a Full English! I wouldn't stay this hotel has been any better than other hotels I've stayed in when it comes to cleanliness of the hotel but it is a good hotel and I would recommend to all visiting Central London!

Stayed: March 2017, traveled with family

Value
Sleep Quality

Service

Ask Liolol about Radisson Blu Edwardian Mercer Street Hotel

#10. Legal

IVA
Active on Messenger

Yes

Are you planning to study in Canada for more than six months?

Yes

In Canada, the steps to get a study permit sometimes vary depending on where you plan to study. Are you planning to study in the Province of Quebec or elsewhere in Canada?

Any other province

Do you have a letter of admission from your institution?

Yes

Seems like you're ready to apply !

If you want to check our service packages right now, please press "Book". If you already know what kind of help you need from us, choose "Apply for a quote". We'll be in touch with you shortly. You can also check our free Student Visa Guide !

Book
Student Visa Guide
Get a Quote

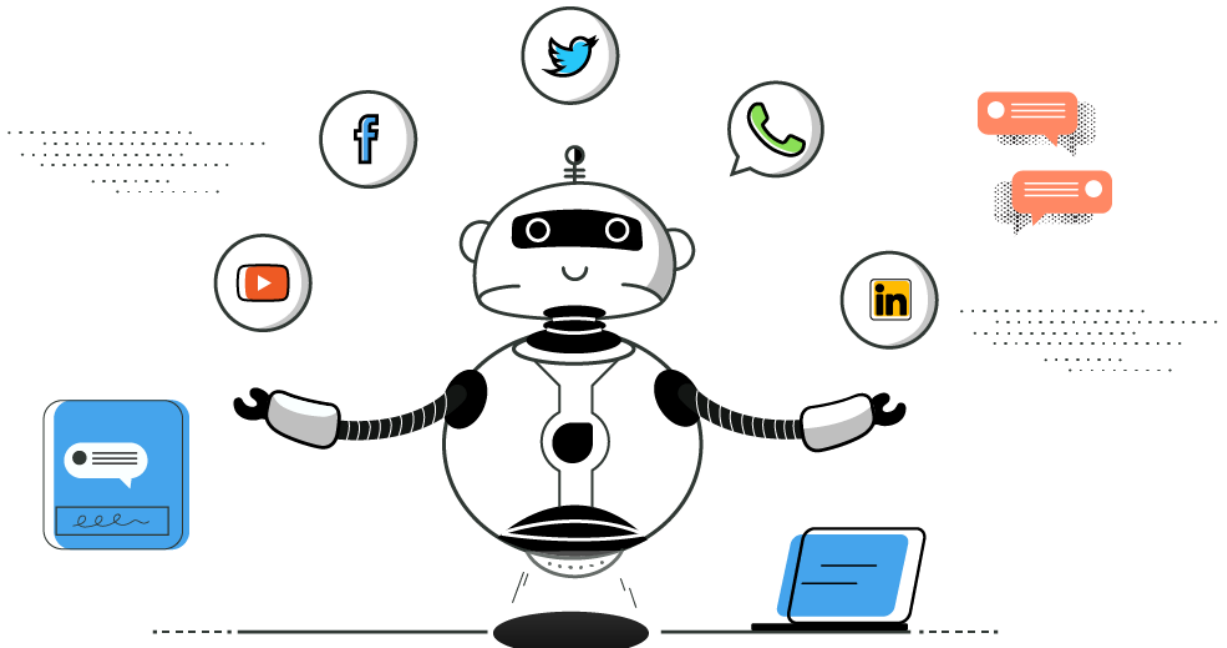
Sp

Chatbots have also shown their practicality in the Legal industry as well, they are a great way of optimizing work processes to save both lawyers and their clients time and money. Montreal-based law firm EXEO launched a smart virtual assistant called IVA to help future immigrants. IVA (Immigration Virtual Assistant) makes the process of immigration much easier and less stressful.

IVA is accessible via Facebook Messenger and provides users with information regarding opportunities available for them such as; different immigration program categories, work permit applications, student visas, and permanent residence programs.

11. Future of Chatbots and AI

- Future of chatbots



The future of chatbots is promising, as advances in natural language processing, artificial intelligence, and machine learning continue to make chatbots more effective and useful. Here are some of the potential developments in the future of chatbots:

Improved Natural Language Processing: Natural language processing (NLP) is the backbone of chatbots. As NLP technology continues to improve, chatbots will be able to understand and interpret human language more accurately and respond more effectively.

Multi-lingual Capabilities: Chatbots will be able to converse in multiple languages, making them more accessible to a global audience.

Advanced Machine Learning: Machine learning algorithms will become more advanced, allowing chatbots to learn from user interactions more quickly and accurately. This will enable chatbots to provide more personalized and relevant responses to users.

Increased Automation: Chatbots will become more autonomous, allowing them to handle more complex tasks without human intervention. This will help to improve efficiency and reduce costs for businesses.

Chatbot

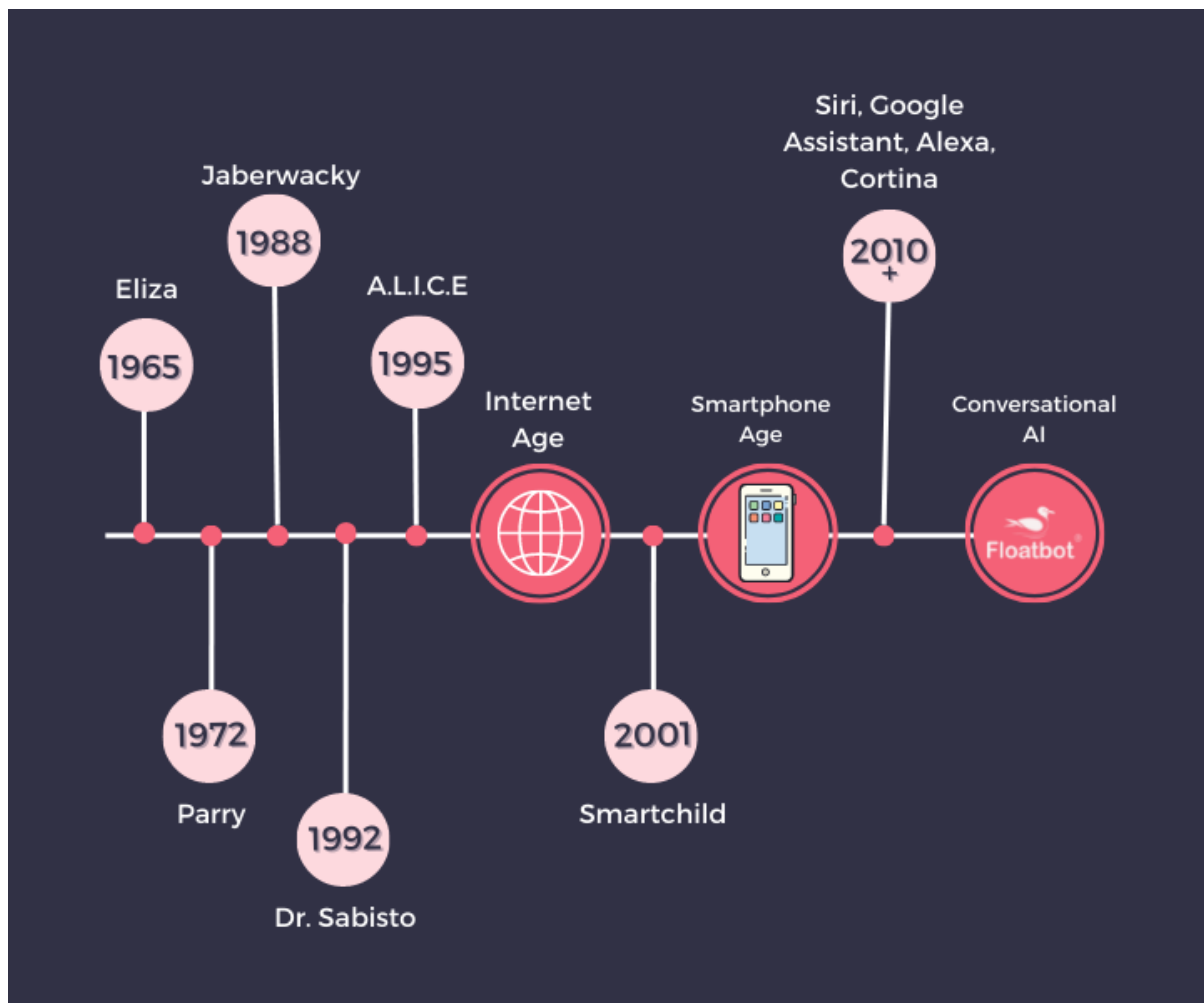
Improved Voice Recognition: Voice recognition technology will continue to improve, allowing chatbots to better understand and respond to spoken input. This will be particularly useful for chatbots used in voice assistants and phone-based customer service applications.

Increased Integration with IoT Devices: Chatbots will be integrated with Internet of Things (IoT) devices, allowing users to control and interact with smart home devices, wearables, and other IoT devices using voice or text commands.

Emotional Intelligence: Chatbots will be designed with emotional intelligence, allowing them to recognize and respond to user emotions. This will improve user engagement and satisfaction.

Overall, chatbots will continue to evolve and become more advanced, leading to better user experiences and improved business outcomes. As chatbots become more sophisticated, they will become a more integral part of our daily lives.

- Advancements in AI technology





Chatbot

There have been significant advancements in artificial intelligence (AI) technology for chatbots in recent years, enabling them to provide more natural and engaging conversation experiences. Here are some of the key advancements in AI technology for chatbots:

Natural Language Processing (NLP): NLP technology has improved significantly, allowing chatbots to better understand and interpret human language. This includes improvements in language models, sentiment analysis, and entity recognition.

Machine Learning (ML): Machine learning algorithms have become more sophisticated, enabling chatbots to learn from user interactions and provide more accurate and relevant responses. This includes advancements in deep learning, neural networks, and reinforcement learning.

Conversational AI: Conversational AI technologies enable chatbots to engage in more natural and human-like conversations with users. This includes advancements in natural language understanding, context management, and dialog management.

Voice Recognition: Voice recognition technology has improved, enabling chatbots to better understand and respond to spoken input. This includes advancements in speech recognition, speaker recognition, and text-to-speech.

Personalization: Chatbots can now be personalized to individual users, enabling them to provide more relevant and personalized responses. This includes advancements in recommendation systems, user profiling, and personalization algorithms.

Chatbot Analytics: Chatbot analytics tools enable businesses to analyze user interactions with chatbots, enabling them to optimize and improve their chatbot performance. This includes advancements in data analytics, machine learning, and visualization tools.

Overall, these advancements in AI technology are making chatbots more effective and useful for businesses and consumers. As AI technology continues to evolve, we can expect even more advancements in chatbot capabilities and functionality, enabling them to become even more integrated into our daily lives.



12. Conclusion

Chatbots are becoming increasingly popular due to their ability to provide efficient and personalized interactions with customers. With advancements in artificial intelligence technology, chatbots are becoming more sophisticated and capable of understanding and responding to natural language. They can be integrated with various platforms and devices, allowing businesses to engage with their customers on multiple channels.

The future of chatbots is bright, with the potential for even more advanced features such as emotional intelligence and greater automation. As businesses continue to adopt chatbots to enhance their customer service and engagement, we can expect chatbots to become an even more integral part of our daily lives.

The main objectives of the project were to develop an algorithm that will be used to identify answers related to user submitted questions. To develop a database where all the related data will be stored and to develop a web interface. The web interface developed had two parts, one for simple users and one for the administrator.

Background research took place, which included an overview of the conversation procedure and any relevant chat bots available. A database was developed, which stores information about questions, answers, keywords, logs and feedback messages. A usable system was designed, developed and deployed to the web server on two occasions. An evaluation took place from data collected by potential students of the University. Also, after received feedback from the first deployment, extra requirements were introduced and implemented



Chatbot

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